

MODEL RUN TRANSCRIPT

Generated: 2026-08-12T10:03:07

Source: data/json/noise_experiments/v2/noise2i_full8match_game/claude-sonnet-4.5/game/noisy_bidirectional_informed_memprimary_full8match/noise2i_full8match_game_002_neutral_rep02.json

RUN METADATA

```
model: anthropic/claude-sonnet-4.5
temperature: 0.8
num_turns: 10
num_agents: 2
memory_capacity: 6
myth_topic_id: anything
myth_topic: anything
game_params_name: noisy_bidirectional_informed_memprimary_full8match
noise_config: {
  "applies_to": "both",
  "inform_agents": true,
  "range": 1.0,
  "type": "uniform"
}
other_player_names: default
replicate_id: 2
myth_default_prompt_key: myth_writing_default
myth_later_prompt_key: myth_writing_later_rounds
history_policy: self_and_coplayer
self_history_window: 0
coplayer_history_window: 3
show_agent_names: false
defector_ratio_requested: 0.0
defector_ratio_actual: 0.0
defector_count: 0
defector_agent_ids: []
defector_seed: 2
task_order: [
  "game"
]
agent_names: hidden in transcript labels because show_agent_names is false
agent_types: {
  "Agent_1": "standard",
  "Agent_2": "standard"
}
```

Coverage: exact chat message snapshots sent to the model are included for every recorded LLM call.

AGENTS

Agent_1; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Agent_2; model=anthropic/claude-sonnet-4.5; memory_capacity=6

System prompt:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

=====

ROUND 1

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$3; payoffs Agent_1=\$5, Agent_2=\$6;

noise/visible: sent visible=\$2.68, received visible=\$8.04, returned visible=\$2.22

Action validation: {"returned": {"amount": 3.0, "clamped": false, "max": 9.0, "min": 0, "raw": 3.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 5.0,

"Agent_2": 6.0

}

Visible balances after round: {

"Agent_1": 4.220000000000001,

"Agent_2": 5.040000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:58:54

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 1,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Assistant output:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

Reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actually sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

```
Usage: {
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_1",
      "Agent_2"
    ],
    "investor": "Agent_1",
```

```

    "roles": {
      "Agent_1": "investor",
      "Agent_2": "trustee"
    },
    "trustee": "Agent_2"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 1,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to The sender? (0-8.040000000000001)

Assistant output:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- I received \$8.04 after the 3x multiplication
- The sender currently has \$2.32 remaining

To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures both of us benefit from the cooperation
3. Encourages future cooperation

If I return \$3.00:

- Sender ends with: \$2.32 + \$3.00 = \$5.32
- I end with: \$8.04 - \$3.00 = \$5.04
- Both of us gain from the exchange

This seems fair and reciprocates their generous trust.

```
{'return': 3.0}
```

Reasoning:

Let me think about this carefully.

As the RECEIVER:

- The sender sent \$2.68 (53.6% of their \$5 endowment)
- I received \$8.04 (which is $\$2.68 \times 3$)
- I need to decide how much to return (between \$0 and \$8.04)

The sender showed trust by sending more than half of their endowment. Let me consider what would be fair:

- The sender kept: $\$5 - \$2.68 = \$2.32$
- I received: \$8.04
- Total wealth created: $\$2.32 + \$8.04 = \$10.36$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total wealth would be: \$5.18 each
- For the sender to have \$5.18, they need: $\$5.18 - \$2.32 = \$2.86$ returned
- This would leave me with: $\$8.04 - \$2.86 = \$5.18$

Alternatively, I could return approximately what they sent, which was \$2.68. This would give:

- Sender final: $\$2.32 + \$2.68 = \$5.00$
- Me final: $\$8.04 - \$2.68 = \$5.36$

Or I could think about proportional reciprocation. They sent 53.6% of their endowment. If I return a similar proportion:

- 53.6% of \$8.04 = \$4.31
- Sender final: $\$2.32 + \$4.31 = \$6.63$
- Me final: $\$8.04 - \$4.31 = \$3.73$

Given that the sender showed significant trust (sending more than half), I should reciprocate generously. The fair 50-50 split seems reasonable, or I could be even more generous to encourage future cooperation.

...

```
Usage: {  
  "input_tokens": 402,  
  "output_tokens": 737,  
  "reasoning_tokens": 0  
}
```

ROUND 2

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.7; received \$8.1; Agent_1 returned \$0; payoffs Agent_2=\$2.3, Agent_1=\$8.1; noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.44

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 8.100000000000001, "min": 0, "raw": 0.0}, "sent": {"amount": 2.7, "clamped": false, "max": 5, "min": 0, "raw": 2.7}}

Balances after round: {

"Agent_1": 13.100000000000001,
"Agent_2": 8.3

}

Visible balances after round: {

"Agent_1": 4.220000000000001,
"Agent_2": 7.780000000000001

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:20

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 2,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to The sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)

1. Matches the trust level shown to me
2. Creates potential for mutual benefit
3. Encourages ongoing cooperation

Sending approximately 54% of my endowment:

```
{'send': 2.7}
```

Reasoning:

Now I'm the sender in Round 2. Let me think about this:

From Round 1:

- I was the receiver
- The sender sent me \$2.68 (53.6% of their endowment)
- I received \$8.04
- I returned \$3.00
- I ended with \$5.04

My total earnings are \$5.04, which matches what I calculated.

Now I'm the sender:

- I have \$5 this round
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

The previous sender showed good trust by sending 53.6% of their endowment. I returned \$3.00 out of \$8.04, which is about 37.3% of what I received.

To build reciprocity and cooperation, I should also show trust. The previous sender sent 53.6%, which was \$2.68. If I send a similar proportion:

- 53.6% of \$5 = \$2.68

Or I could send a similar absolute amount. The sender sent \$2.68.

Given that:

1. The previous sender showed significant trust
2. I benefited from that trust
3. Cooperation generally leads to better outcomes for both parties
4. I want to encourage continued cooperation

I should send a substantial amount to demonstrate trust and reciprocity. Sending around \$2.50-\$3 seems reasonable.

Let me send \$2.70 (54% of my endowment), which is very similar to what the sender did in Round 1. This shows I'm reciprocating their trust level.

Actually, let me reconsider. The note says there may be communication noise. The sender might have actually sent a different amount than what I saw. But I should base my decision on what I observed.

I'll send \$2.70 to match the trust level shown to me.

```
Usage: {
  "input_tokens": 637,
  "output_tokens": 539,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:34

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
  },
```

```

    "investor": "Agent_2",
    "roles": {
        "Agent_1": "trustee",
        "Agent_2": "investor"
    },
    "trustee": "Agent_1"
},
"role": "trustee",
"role_label": "Receiver",
"round": 2,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actually sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
```json
{"return": 0}
```
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

Reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept $5 - 3 = 2$ initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

Wait, let me recalculate. If the sender sent me \$3 in Round 1, that became \$9. If I returned \$2.22, then:

- Sender's final: $5 - 3 + 2.22 = 4.22$?
- My final from Round 1: $9 - 2.22 = 6.78$

But the problem says there may be communication noise. So maybe what I returned wasn't what they saw, or what they sent wasn't what I received? This could explain the lack of trust in Round 2.

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

```
Usage: {
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

ROUND 3

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2; received \$6; Agent_2 returned \$0; payoffs Agent_1=\$3, Agent_2=\$6;
noise/visible: sent visible=\$0, received visible=\$0
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 6.0, "min": 0, "raw": 0.0}, "sent": {"amount": 2.0, "clamped": false, "max": 5, "min": 0, "raw": 2.0}}

Balances after round: {

"Agent_1": 16.1,
"Agent_2": 14.3

}

Visible balances after round: {

"Agent_1": 7.2200000000000001,
"Agent_2": 7.7800000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T09:59:49

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"

```

],
"investor": "Agent_1",
"roles": {
  "Agent_1": "investor",
  "Agent_2": "trustee"
},
"trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 3,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```

'''json
{"send": 3}

```

'''

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actually sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what t

reasoning:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept $\$5 - \$3 = \$2$ initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

usage:

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

content:

Assistant output:

Reasoning:

Page 15 of 112

Round 1: I was the SENDER and sent \$3.
- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22
- So the receiver returned about \$2.22 to me

Round 2: I was the RECEIVER
- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

Round 3: I'm the SENDER again
- I have \$5 to potentially send
- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

```
Usage: {  
  "input_tokens": 787,  
  "output_tokens": 675,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T09:59:52

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 3,  
  "task": "game",
```



```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 1: You are the RECEIVER. The sender sent $2.68, that is 53.6% of their total endowme
    nt. You received $8.040000000000001. How much do you return to The sender? (0-8.04000000
    0000001)

Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed significant trust by sending $2.68 (53.6% of their endowment)
  - I received $8.04 after the 3x multiplication
  - The sender currently has $2.32 remaining

  To reciprocate this trust fairly, I should return an amount that:
  1. Rewards their trust
  2. Ensures both of us benefit from the cooperation
  3. Encourages future cooperation

  If I return $3.00:
  - Sender ends with: $2.32 + $3.00 = $5.32
  - I end with: $8.04 - $3.00 = $5.04
  - Both of us gain from the exchange

  This seems fair and reciprocates their generous trust.

  {'return': 3.0}
reasoning:
  Let me think about this carefully.

  As the RECEIVER:
  - The sender sent $2.68 (53.6% of their $5 endowment)

```


- I have \$5 this round
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

The previous sender showed good trust by sending 53.6% of their endowment. I returned \$3.00 out of \$8.04, which is about 37.3% of what I received.

To build reciprocity and cooperation, I should also show trust. The previous sender sent 53.6%, which was \$2.68. If I send a similar proportion:

- 53.6% of \$5 = \$2.68

Or I could send a similar absolute amount. The sender sent \$2.68.

Given that:

1. The previous sender showed significant trust
2. I benefited from that trust
3. Cooperation generally leads to better outcomes for both parties
4. I want to encourage continued cooperation

I should send a substantial amount to demonstrate trust and reciprocity. Sending around \$2.50-\$3 seems reasonable.

Let me send \$2.70 (54% of my endowment), which is very similar to what the sender did in Round 1. This shows I'm reciprocating their trust level.

Actually, let me reconsider. The note says there may be communication noise. The sender might have actually sent a different amount than what I saw. But I should base my decision on what I observed.

I'll send \$2.70 to match the trust level shown to me.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 539,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.780000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, there is nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, there is nothing to return. The only valid response is to return \$0.

The format should be: {'return': 0}

This is straightforward - I can't return what I didn't receive.

Usage: {

```
"input_tokens": 806,
"output_tokens": 110,
"reasoning_tokens": 0
}
```

=====

ROUND 4

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$2.5; received \$7.5; Agent_1 returned \$0.4; payoffs Agent_2=\$2.9,

```

Agent_1=$7.1; noise/visible: sent visible=$0.29, received visible=$0.87, returned visible=$0
Action validation: {"returned": {"amount": 0.4, "clamped": false, "max": 7.5, "min": 0, "raw":
0.4}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}
Balances after round: {
  "Agent_1": 23.200000000000003,
  "Agent_2": 17.2
}
Visible balances after round: {
  "Agent_1": 7.69,
  "Agent_2": 10.280000000000001
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:08

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```

  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 4,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may al
so be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths

, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to The sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- I received \$8.04 after the 3x multiplication
- The sender currently has \$2.32 remaining

To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures both of us benefit from the cooperation
3. Encourages future cooperation

If I return \$3.00:

- Sender ends with: $\$2.32 + \$3.00 = \$5.32$
- I end with: $\$8.04 - \$3.00 = \$5.04$
- Both of us gain from the exchange

This seems fair and reciprocates their generous trust.

{'return': 3.0}

reasoning:

Let me think about this carefully.

As the RECEIVER:

- The sender sent \$2.68 (53.6% of their \$5 endowment)
- I received \$8.04 (which is $\$2.68 \times 3$)
- I need to decide how much to return (between \$0 and \$8.04)

The sender showed trust by sending more than half of their endowment. Let me consider what would be fair:

- The sender kept: $\$5 - \$2.68 = \$2.32$
- I received: \$8.04
- Total wealth created: $\$2.32 + \$8.04 = \$10.36$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total wealth would be: \$5.18 each
- For the sender to have \$5.18, they need: $\$5.18 - \$2.32 = \$2.86$ returned
- This would leave me with: $\$8.04 - \$2.86 = \$5.18$

Alternatively, I could return approximately what they sent, which was \$2.68. This would give:

- Sender final: $\$2.32 + \$2.68 = \$5.00$
- Me final: $\$8.04 - \$2.68 = \$5.36$

Or I could think about proportional reciprocation. They sent 53.6% of their endowment. If I return a similar proportion:

- 53.6% of \$8.04 = \$4.31
- Sender final: $\$2.32 + \$4.31 = \$6.63$
- Me final: $\$8.04 - \$4.31 = \$3.73$

Given that the sender showed significant trust (sending more than half), I should reciprocate generously. The fair 50-50 split seems reasonable, or I could be even more generous to encourage future cooperation.

...

usage:

Message 4 role=user
content:
Round 2

How much do you send to the receiver this round? (0-5)

Looking at the previous round, the sender demonstrated significant trust by sending 53.6% of their endowment, and I benefited from that cooperation.

1. Matches the trust level shown to me
2. Creates potential for mutual benefit
3. Encourages ongoing cooperation

```
{'send': 2.7}
```

Now I'm the sender in Round 2. Let me think about this:

- I was the receiver
- The sender sent me \$2.68 (53.6% of their endowment)
- I received \$8.04
- I returned \$3.00
- I ended with \$5.04

- I have \$5 this round
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

To build reciprocity and cooperation, I should also show trust. The previous sender sent 53.6%, which was \$2.68. If I send a similar proportion:

- 53.6% of \$5 = \$2.68

1. The previous sender showed significant trust
2. I benefited from that trust
3. Cooperation generally leads to better outcomes for both parties
4. I want to encourage continued cooperation

Actually, let me reconsider. The note says there may be communication noise. The sender might have actually sent a different amount than what I saw. But I should base my decision on what I observed.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 539,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.780000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, there is nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, there is nothing to return. The only valid response is to return \$0.

The format should be: {'return': 0}

This is straightforward - I can't return what I didn't receive.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 110,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.780000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern so far:

Round 1: As receiver, the sender trusted me with 53.6% - I returned \$3.00

Round 2: As sender, I sent 54% - the receiver returned generously

Round 3: As receiver, the sender sent \$0 - possibly due to communication noise mentioned in the instructions

Given that:

1. Previous cooperation rounds were successful with good returns
2. The \$0 in round 3 might be communication noise rather than intentional
3. Maintaining cooperation has been beneficial overall

I'll continue with a cooperative strategy and send approximately 50%:

{'send': 2.5}

Reasoning:

Let me analyze what happened:

Round 1: I was receiver

- Sender sent \$2.68 (53.6%)
- I received \$8.04
- I returned \$3.00
- Sender ended with \$5.32, I ended with \$5.04

Round 2: I was sender

- I sent \$2.7 (54%)
- Receiver would have received \$8.10
- My total earnings after round 2: \$7.78
- This means I started round 2 with \$5.04, sent \$2.7, so I had \$2.34 left

- My total is now \$7.78, which means the receiver returned $\$7.78 - \$2.34 = \$5.44$
- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

```
Usage: {
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:25

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 4,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.

- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actually sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept \$5 - \$3 = \$2 initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

- Sender's final: $\$5 - \$3 + \$2.22 = \4.22 ?
- My final from Round 1: $\$9 - \$2.22 = \$6.78$

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Round 3

How much do you send to the receiver this round? (0-5)

Looking at the situation:

- Round 1: I sent \$3, and based on my total of \$4.22, I received back approximately \$2.22 ($\$5 - \$3 + \$2.22 = \4.22)
- This means the receiver kept most of the \$9 they received (about 75%)
- Round 2: As receiver, I got \$0 from a different sender

1. The receiver wasn't very cooperative, OR
2. Communication noise affected what they saw/returned

```
'''json
{"send": 2}
'''
```

Let me think about what happened:

- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22
- So the receiver returned about \$2.22 to me

- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

- I have \$5 to potentially send
- My current total is \$4.22

Page 27 of 112

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Assistant output:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

Reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3

- I kept: $\$5 - \$2 + \text{returned} = \$7.22$

- So: $\$3 + \text{returned} = \7.22

- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3
- If I received back \$2.22, I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0
- I gained: $\$0 - \$0 = \$0$
- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2
- Total before Round 3: \$4.22
- Total after Round 3: \$7.22
- Change: \$3.00
- I kept: $\$5 - \$2 + \text{returned} = \$3$
- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87
- The sender sent \$0.29, which became \$0.87 after 3x multiplier
- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

```
Usage: {  
  "input_tokens": 1098,  
  "output_tokens": 843,  
  "reasoning_tokens": 0  
}
```

ROUND 5

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$1; received \$3; Agent_2 returned \$0.15; payoffs Agent_1=\$4.15, Agent_2=\$2.85; noise/visible: sent visible=\$0.13, received visible=\$0.39, returned visible=\$0.81
Action validation: {"returned": {"amount": 0.15, "clamped": false, "max": 3.0, "min": 0, "raw": 0.15}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 27.35,
"Agent_2": 20.05

}

Visible balances after round: {

"Agent_1": 12.5,
"Agent_2": 10.520000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:00:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"

```

},
"role": "investor",
"role_label": "Sender",
"round": 5,
"task": "game",
"task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```

'''json
{"send": 3}
'''

```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategy.

tegitically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actually sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept \$5 - \$3 = \$2 initially

4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

Wait, let me recalculate. If the sender sent me \$3 in Round 1, that became \$9. If I returned \$2.22, then:

- Sender's final: $\$5 - \$3 + \$2.22 = \4.22 ?
- My final from Round 1: $\$9 - \$2.22 = \$6.78$

But the problem says there may be communication noise. So maybe what I returned wasn't what they saw, or what they sent wasn't what I received? This could explain the lack of trust in Round 2.

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.22000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

```
**Analysis:**
```

- Round 1: I sent \$3, and based on my total of \$4.22, I received back approximately \$2.22 ($\$5 - \$3 + \$2.22 = \4.22)
- This means the receiver kept most of the \$9 they received (about 75%)
- Round 2: As receiver, I got \$0 from a different sender

The relatively low return rate in Round 1 suggests either:

1. The receiver wasn't very cooperative, OR
2. Communication noise affected what they saw/returned

I'll send a smaller amount this round to test if cooperation can be rebuilt while limiting downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. If they return \$2-3, we both benefit. If they don't reciprocate well again, I've limited my loss. This allows me to gauge whether the pattern continues or if Round 1 was anomalous.

reasoning:

Let me think about what happened:

Round 1: I was the SENDER and sent \$3.

- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22

- So the receiver returned about \$2.22 to me

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

Round 3: I'm the SENDER again

- I have \$5 to potentially send
- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? ($0 - 0.8699999999999999$)

Message 9 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22
- I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0
- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22
- Change from \$4.22 to \$7.22 = \$3
- I kept: $\$5 - \$2 + \text{returned} = \$7.22$
- So: $\$3 + \text{returned} = \7.22
- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3
- If I received back \$2.22, I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0
- I gained: $\$0 - \$0 = \$0$
- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2
- Total before Round 3: \$4.22
- Total after Round 3: \$7.22
- Change: \$3.00
- I kept: $\$5 - \$2 + \text{returned} = \$3$
- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87
- The sender sent \$0.29, which became \$0.87 after 3x multiplier
- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
  "input_tokens": 1098,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.
How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)
- Round 3: Sent \$2, got back \$0 (receiver kept everything!)
- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes
- Each round might have different partners
- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

```
'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. I

f they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

Reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5
- Sent \$3, so I had \$2 left
- My total after Round 1 was \$4.22
- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round
- Started this round with \$5, sent \$2, so I had \$3 left
- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$
- After Round 3, my total was \$7.22
- So: $\$4.22 + \$3 + \text{returned} = \$7.22$
- Returned = $\$7.22 - \$7.22 = \$0$
- Wait, that doesn't work...
- Actually, let me recalculate. After Round 2, I had \$4.22
- In Round 3, I start fresh with \$5, send \$2, keep \$3
- So I have \$3 + whatever is returned
- After Round 3, my TOTAL is \$7.22
- So $\$4.22 (\text{previous}) + \$3 (\text{kept in Round 3}) + \text{returned} = \7.22
- Returned = \$0

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40
- So I kept $\$0.87 - \$0.40 = \$0.47$
- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?
- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

Usage: {

```
"input_tokens": 1340,  
"output_tokens": 902,  
"reasoning_tokens": 0
```

}

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:00:55

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",  
"dyad_id": "dyad_1",  
"move_index": 1,  
"opponent_id": "Agent_1",  
"pairing": {  
  "agents": [  
    "Agent_1",  
    "Agent_2"  
  ],  
  "investor": "Agent_1",  
  "roles": {  
    "Agent_1": "investor",  
    "Agent_2": "trustee"  
  },  
  "trustee": "Agent_2"  
},  
"role": "trustee",  
"role_label": "Receiver",  
"round": 5,  
"task": "game",
```

```

"task_index": 0
}
Exact input messages sent to model:
Message 1 role=system
content:
  You are an agent participating in a multi-task experiment. You will play a game and may al
    so be asked to write myths.

  GAME:
  You will play different roles across multiple rounds.

  When you are the SENDER:
  - You receive $5 each round
  - You decide how much to SEND to the RECEIVER ($0-$5)
  - Amount sent is multiplied by 3x
  - Receiver decides how much to RETURN
  - You keep: $5 - sent + returned
  - Receiver keeps: received - returned
  - Response format: {'send': <amount>}

  When you are the RECEIVER:
  - Sender receives $5
  - Sender decides how much to SEND to you ($0-$5)
  - Amount sent is multiplied by 3x
  - You decide how much to RETURN
  - Sender keeps: $5 - sent + returned
  - You keep: received - returned
  - Response format: {'return': <amount>}

  MYTH WRITING:
  You may also be asked to write short myths (stories) on various topics. When writing myths
    , follow the instructions provided in each prompt.

  IMPORTANT: Provide your decision in the correct JSON format.

  NOTE: There may be communication noise in this game.
  The amounts you see may differ slightly from what was actually sent/returned.
  This is part of the experimental design.

Message 2 role=user
content:
  Round 1: You are the RECEIVER. The sender sent $2.68, that is 53.6% of their total endowme
    nt. You received $8.040000000000001. How much do you return to The sender? (0-8.04000000
    0000001)

Message 3 role=assistant
content:
  Looking at this situation:

  - The sender showed significant trust by sending $2.68 (53.6% of their endowment)
  - I received $8.04 after the 3x multiplication
  - The sender currently has $2.32 remaining

  To reciprocate this trust fairly, I should return an amount that:
  1. Rewards their trust
  2. Ensures both of us benefit from the cooperation
  3. Encourages future cooperation

  If I return $3.00:
  - Sender ends with: $2.32 + $3.00 = $5.32
  - I end with: $8.04 - $3.00 = $5.04
  - Both of us gain from the exchange

  This seems fair and reciprocates their generous trust.

  {'return': 3.0}
reasoning:
  Let me think about this carefully.

  As the RECEIVER:
  - The sender sent $2.68 (53.6% of their $5 endowment)

```

- I received \$8.04 (which is $\$2.68 \times 3$)
- I need to decide how much to return (between \$0 and \$8.04)

The sender showed trust by sending more than half of their endowment. Let me consider what would be fair:

- The sender kept: $\$5 - \$2.68 = \$2.32$
- I received: $\$8.04$
- Total wealth created: $\$2.32 + \$8.04 = \$10.36$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total wealth would be: \$5.18 each
- For the sender to have \$5.18, they need: $\$5.18 - \$2.32 = \$2.86$ returned
- This would leave me with: $\$8.04 - \$2.86 = \$5.18$

Alternatively, I could return approximately what they sent, which was \$2.68. This would give:

- Sender final: $\$2.32 + \$2.68 = \$5.00$
- Me final: $\$8.04 - \$2.68 = \$5.36$

Or I could think about proportional reciprocation. They sent 53.6% of their endowment. If I return a similar proportion:

- 53.6% of \$8.04 = \$4.31
- Sender final: \$2.32 + \$4.31 = \$6.63
- Me final: \$8.04 - \$4.31 = \$3.73

Given that the sender showed significant trust (sending more than half), I should reciprocate generously. The fair 50-50 split seems reasonable, or I could be even more generous to encourage future cooperation.

• • •

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 737,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the SENDER this round. Your total earnings across all rounds are \$5.040000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the previous round, the sender demonstrated significant trust by sending 53.6% of their endowment, and I benefited from that cooperation.

To reciprocate that trust and encourage continued cooperation, I should send a similar proportion. This strategy:

1. Matches the trust level shown to me
2. Creates potential for mutual benefit
3. Encourages ongoing cooperation

Sending approximately 54% of my endowment:

{ 'send' : 2.7 }

reasoning:

Now I'm the sender in Round 2. Let me think about this:

From Round 1:

- I was the receiver
- The sender sent me \$2.68 (53.6% of their endowment)
- I received \$8.04
- I returned \$3.00
- I ended with \$5.04

My total earnings are \$5.04, which matches what I calculated.

Now I'm the sender:

- I have \$5 this round
- Whatever I send gets multiplied by 3x
- The receiver decides how much to return

The previous sender showed good trust by sending 53.6% of their endowment. I returned \$3.00 out of \$8.04, which is about 37.3% of what I received.

To build reciprocity and cooperation, I should also show trust. The previous sender sent 53.6%, which was \$2.68. If I send a similar proportion:

- 53.6% of \$5 = \$2.68

Or I could send a similar absolute amount. The sender sent \$2.68.

Given that:

1. The previous sender showed significant trust
2. I benefited from that trust
3. Cooperation generally leads to better outcomes for both parties
4. I want to encourage continued cooperation

I should send a substantial amount to demonstrate trust and reciprocity. Sending around \$2.50-\$3 seems reasonable.

Let me send \$2.70 (54% of my endowment), which is very similar to what the sender did in Round 1. This shows I'm reciprocating their trust level.

Actually, let me reconsider. The note says there may be communication noise. The sender might have actually sent a different amount than what I saw. But I should base my decision on what I observed.

I'll send \$2.70 to match the trust level shown to me.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 539,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.780000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

Since the sender sent \$0 and I received \$0, there is nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, there is nothing to return. The only valid response is to return \$0.

The format should be: {'return': 0}

This is straightforward - I can't return what I didn't receive.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 110,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.780000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern so far:

Round 1: As receiver, the sender trusted me with 53.6% - I returned \$3.00

Round 2: As sender, I sent 54% - the receiver returned generously

Round 3: As receiver, the sender sent \$0 - possibly due to communication noise mentioned in the instructions

Given that:

1. Previous cooperation rounds were successful with good returns
2. The \$0 in round 3 might be communication noise rather than intentional
3. Maintaining cooperation has been beneficial overall

I'll continue with a cooperative strategy and send approximately 50%:

{'send': 2.5}

reasoning:

Let me analyze what happened:

Round 1: I was receiver

- Sender sent \$2.68 (53.6%)
- I received \$8.04
- I returned \$3.00
- Sender ended with \$5.32, I ended with \$5.04

Round 2: I was sender

- I sent \$2.7 (54%)
- Receiver would have received \$8.10
- My total earnings after round 2: \$7.78
- This means I started round 2 with \$5.04, sent \$2.7, so I had \$2.34 left
- My total is now \$7.78, which means the receiver returned \$7.78 - \$2.34 = \$5.44
- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Assistant output:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

{'return': 0.15}

Reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return something to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

```
Usage: {  
  "input_tokens": 1099,  
  "output_tokens": 513,  
  "reasoning_tokens": 0  
}
```

=====

ROUND 6

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1.5; received \$4.5; Agent_1 returned \$0; payoffs Agent_2=\$3.5, Agent_1=\$4.5;
noise/visible: sent visible=\$0, received visible=\$0, returned visible=\$0.14
Action validation: {"returned": {"amount": 0, "clamped": false, "max": 4.5, "min": 0, "raw":


```

    0.0}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}
Balances after round: {
  "Agent_1": 31.85,
  "Agent_2": 23.55
}
Visible balances after round: {
  "Agent_1": 12.5,
  "Agent_2": 14.160000000000002
}

```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:16
 Model: anthropic/claude-sonnet-4.5
 Temperature: 0.8

```

Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 0,
  "opponent_id": "Agent_1",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "investor",
  "role_label": "Sender",
  "round": 6,
  "task": "game",
  "task_index": 0
}

```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to The sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- I received \$8.04 after the 3x multiplication
- The sender currently has \$2.32 remaining

To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures both of us benefit from the cooperation
3. Encourages future cooperation

If I return \$3.00:

- Sender ends with: $\$2.32 + \$3.00 = \$5.32$
- I end with: $\$8.04 - \$3.00 = \$5.04$
- Both of us gain from the exchange

This seems fair and reciprocates their generous trust.

{'return': 3.0}

reasoning:

Let me think about this carefully.

As the RECEIVER:

- The sender sent \$2.68 (53.6% of their \$5 endowment)
- I received \$8.04 (which is $\$2.68 \times 3$)
- I need to decide how much to return (between \$0 and \$8.04)

The sender showed trust by sending more than half of their endowment. Let me consider what would be fair:

- The sender kept: $\$5 - \$2.68 = \$2.32$
- I received: \$8.04
- Total wealth created: $\$2.32 + \$8.04 = \$10.36$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total wealth would be: \$5.18 each
- For the sender to have \$5.18, they need: $\$5.18 - \$2.32 = \$2.86$ returned
- This would leave me with: $\$8.04 - \$2.86 = \$5.18$

Alternatively, I could return approximately what they sent, which was \$2.68. This would give:

- Sender final: $\$2.32 + \$2.68 = \$5.00$
- Me final: $\$8.04 - \$2.68 = \$5.36$

Or I could think about proportional reciprocation. They sent 53.6% of their endowment. If

I return a similar proportion:

- 53.6% of \$8.04 = \$4.31
- Sender final: $\$2.32 + \$4.31 = \$6.63$
- Me final: $\$8.04 - \$4.31 = \$3.73$

Given that the sender showed significant trust (sending more than half), I should reciprocate generously. The fair 50-50 split seems reasonable, or I could be even more generous to encourage future cooperation.

...

usage:

```
{
  "input_tokens": 402,
```

```

        "output_tokens": 737,
        "reasoning_tokens": 0
    }
Message 4 role=user
  content:
    Round 2

    You are the SENDER this round. Your total earnings across all rounds are $5.040000000000000
    1.
    How much do you send to the receiver this round? (0-5)
Message 5 role=assistant
  content:
    Looking at the previous round, the sender demonstrated significant trust by sending 53.6%
    of their endowment, and I benefited from that cooperation.

    To reciprocate that trust and encourage continued cooperation, I should send a similar pro
    portion. This strategy:
    1. Matches the trust level shown to me
    2. Creates potential for mutual benefit
    3. Encourages ongoing cooperation

    Sending approximately 54% of my endowment:

    {'send': 2.7}
  reasoning:
    Now I'm the sender in Round 2. Let me think about this:

    From Round 1:
    - I was the receiver
    - The sender sent me $2.68 (53.6% of their endowment)
    - I received $8.04
    - I returned $3.00
    - I ended with $5.04

    My total earnings are $5.04, which matches what I calculated.

    Now I'm the sender:
    - I have $5 this round
    - Whatever I send gets multiplied by 3x
    - The receiver decides how much to return

    The previous sender showed good trust by sending 53.6% of their endowment. I returned $3.0
    0 out of $8.04, which is about 37.3% of what I received.

    To build reciprocity and cooperation, I should also show trust. The previous sender sent 5
    3.6%, which was $2.68. If I send a similar proportion:
    - 53.6% of $5 = $2.68

    Or I could send a similar absolute amount. The sender sent $2.68.

    Given that:
    1. The previous sender showed significant trust
    2. I benefited from that trust
    3. Cooperation generally leads to better outcomes for both parties
    4. I want to encourage continued cooperation

    I should send a substantial amount to demonstrate trust and reciprocity. Sending around $2
    .50-$3 seems reasonable.

    Let me send $2.70 (54% of my endowment), which is very similar to what the sender did in R
    ound 1. This shows I'm reciprocating their trust level.

    Actually, let me reconsider. The note says there may be communication noise. The sender mi
    ght have actually sent a different amount than what I saw. But I should base my decision
    on what I observed.

    I'll send $2.70 to match the trust level shown to me.
  usage:
    {
      "input_tokens": 637,

```

```

    "output_tokens": 539,
    "reasoning_tokens": 0
  }
Message 6 role=user
  content:
    Round 3

    You are the RECEIVER this round. Your total earnings across all rounds are $7.780000000000000001.
    This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 7 role=assistant
  content:
    Since the sender sent $0 and I received $0, there is nothing to return.

    {'return': 0}
  reasoning:
    The sender sent me $0 this round, so I received $0 after the 3x multiplication.

    Since I received nothing, there is nothing to return. The only valid response is to return $0.

    The format should be: {'return': 0}

    This is straightforward - I can't return what I didn't receive.
  usage:
    {
      "input_tokens": 806,
      "output_tokens": 110,
      "reasoning_tokens": 0
    }
Message 8 role=user
  content:
    Round 4

    You are the SENDER this round. Your total earnings across all rounds are $7.780000000000000001.
    How much do you send to the receiver this round? (0-5)
Message 9 role=assistant
  content:
    Looking at the pattern so far:

    Round 1: As receiver, the sender trusted me with 53.6% - I returned $3.00
    Round 2: As sender, I sent 54% - the receiver returned generously
    Round 3: As receiver, the sender sent $0 - possibly due to communication noise mentioned in the instructions

    Given that:
    1. Previous cooperation rounds were successful with good returns
    2. The $0 in round 3 might be communication noise rather than intentional
    3. Maintaining cooperation has been beneficial overall

    I'll continue with a cooperative strategy and send approximately 50%:

    {'send': 2.5}
  reasoning:
    Let me analyze what happened:

    Round 1: I was receiver
    - Sender sent $2.68 (53.6%)
    - I received $8.04
    - I returned $3.00
    - Sender ended with $5.32, I ended with $5.04

    Round 2: I was sender
    - I sent $2.7 (54%)
    - Receiver would have received $8.10
    - My total earnings after round 2: $7.78
    - This means I started round 2 with $5.04, sent $2.7, so I had $2.34 left
    - My total is now $7.78, which means the receiver returned $7.78 - $2.34 = $5.44

```

- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Message 11 role=assistant

content:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

```
{'return': 0.15}
```

reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return something to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

usage:

```
{
  "input_tokens": 1099,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.5200000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at the pattern of returns I've been receiving:

Round 2: I sent \$2.70, received back a small return

Round 4: I sent \$2.50, received back \$0

Despite my cooperative approach, the returns have been disappointing. This could be due to communication noise affecting what receivers see, or strategic choices by receivers.

Given this pattern, I need to be more cautious while still allowing for the possibility of re-establishing cooperation. I'll send a moderate amount to test whether cooperation can improve, but reduce my exposure.

```
{'send': 1.5}
```

Reasoning:

Let me review the pattern:

Round 1: I was receiver. Sender sent \$2.68 (53.6%), I received \$8.04, I returned \$3.00

Round 2: I was sender. I sent \$2.70 (54%), receiver presumably returned well

Round 3: I was receiver. Sender sent \$0 (communication noise?)

Round 4: I was sender. I sent \$2.50 (50%)

Round 5: I was receiver. Sender sent \$0.13 (2.6%), I received \$0.39, I returned \$0.15

My total earnings are \$10.52, which means:

- Round 1: I got \$5.04 (started with \$5, sent \$2.68, got \$3.00 back would be \$5.32... wait that doesn't match)

Let me recalculate from the receiver perspective in Round 1:

- I received \$8.04
- I returned \$3.00
- I kept \$5.04

So my earnings from Round 1 were \$5.04

Round 2: I was sender with \$5 endowment

- I sent \$2.70

- I had \$2.30 left
- The receiver must have returned something to get me to \$7.78 total

\$7.78 - \$5.04 = \$2.74 earnings in round 2

If I sent \$2.70 and kept \$2.30, then I received back... wait

Actually as sender in round 2, I start fresh with \$5

- I sent \$2.70
- I kept \$2.30
- Receiver returned X
- My total from round 2 = \$2.30 + X

Total after round 2 was \$7.78

So the receiver returned only \$0.44 on the \$8.10 they received-surprisingly low given what I sent. Round 3 was a wash with the sender contributing nothing. In Round 4, I sent \$2.50 and kept the other half, earning exactly that amount back with no return, suggesting the receiver kept everything. Round 5 shows minimal reciprocation again: the sender only sent \$0.13 (2.6 % of the tripled amount), and I returned \$0.15 on my \$0.39 received.

I'm noticing a troubling pattern-my generous sends in Rounds 2 and 4 aren't being matched with proportional returns, while I'm getting minimal amounts as a receiver. This could indicate either communication issues distorting the amounts or a lack of reciprocal trust from the other players....

```
Usage: {
  "input_tokens": 1307,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:26

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 6,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actual

lly sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept \$5 - \$3 = \$2 initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

Wait, let me recalculate. If the sender sent me \$3 in Round 1, that became \$9. If I returned \$2.22, then:

- Sender's final: \$5 - \$3 + \$2.22 = \$4.22 ?
- My final from Round 1: \$9 - \$2.22 = \$6.78

But the problem says there may be communication noise. So maybe what I returned wasn't what they saw, or what they sent wasn't what I received? This could explain the lack of trust.

st in Round 2.

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.220000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

```
**Analysis:**
```

- Round 1: I sent \$3, and based on my total of \$4.22, I received back approximately \$2.22 ($\$5 - \$3 + \$2.22 = \4.22)
- This means the receiver kept most of the \$9 they received (about 75%)
- Round 2: As receiver, I got \$0 from a different sender

The relatively low return rate in Round 1 suggests either:

1. The receiver wasn't very cooperative, OR
2. Communication noise affected what they saw/returned

I'll send a smaller amount this round to test if cooperation can be rebuilt while limiting downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. If they return \$2-3, we both benefit. If they don't reciprocate well again, I've limited my loss. This allows me to gauge whether the pattern continues or if Round 1 was anomalous.

reasoning:

Let me think about what happened:

Round 1: I was the SENDER and sent \$3.

- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22
- So the receiver returned about \$2.22 to me

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

Round 3: I'm the SENDER again

- I have \$5 to potentially send
- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 9 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3
- I kept: \$5 - \$2 + returned = \$7.22
- So: \$3 + returned = \$7.22
- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3

- If I received back \$2.22, I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- I gained: \$0 - \$0 = \$0

- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2

- Total before Round 3: \$4.22

- Total after Round 3: \$7.22

- Change: \$3.00

- I kept: $\$5 - \$2 + \text{returned} = \$3$

- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87

- The sender sent \$0.29, which became \$0.87 after 3x multiplier

- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
  "input_tokens": 1098,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)

- Round 3: Sent \$2, got back \$0 (receiver kept everything!)

- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes

- Each round might have different partners

- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

```
'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. If they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5

- Sent \$3, so I had \$2 left

- My total after Round 1 was \$4.22

- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round

- Started this round with \$5, sent \$2, so I had \$3 left

- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$

- After Round 3, my total was \$7.22

- So: $\$4.22 + \$3 + \text{returned} = \$7.22$

- Returned = $\$7.22 - \$4.22 = \$3$

- Wait, that doesn't work...
- Actually, let me recalculate. After Round 2, I had \$4.22
- In Round 3, I start fresh with \$5, send \$2, keep \$3
- So I have \$3 + whatever is returned
- After Round 3, my TOTAL is \$7.22
- So \$4.22 (previous) + \$3 (kept in Round 3) + returned = \$7.22
- Returned = \$0

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40
- So I kept $\$0.87 - \$0.40 = \$0.47$
- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?
- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

usage:

```
{
  "input_tokens": 1340,
  "output_tokens": 902,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

```
'''json
{"return": 0}
'''
```

****Observation:**** This is the second time as a receiver that I've been sent nothing. The lack of sending suggests either very low trust in the system, previous bad experiences with receivers, or the effects of communication noise discouraging cooperation. It's unfortunate as the 3x multiplier means we're both missing out on potential gains.

Reasoning:

The sender sent me \$0 this round, so after the 3x multiplier I received \$0.

Since I received nothing, I can only return \$0.

My total earnings are now \$12.50, which means in Round 5, I must have gained \$4.81 (from \$7.69 to \$12.50). This suggests:

- I sent \$1 in Round 5
- I should have kept \$4 if I sent \$1
- But I gained \$4.81, which means I got back approximately \$0.81

Wait, let me recalculate:

- Round 4 end: \$7.69
- Round 5: I was sender, sent \$1
 - Started with \$5
 - Sent \$1, so kept \$4
 - Plus whatever was returned
- Total gain = $\$12.5 - \$7.69 = \$4.81$
- So I must have been returned \$0.81
- That means: $\$5 - \$1 + \$0.81 = \4.81 ?

The receiver in Round 5 kept most of the \$3 they received (returning only \$0.81 out of \$3).

Now in Round 6, I'm the receiver and got \$0, so I return \$0.

```
Usage: {
  "input_tokens": 1628,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

=====

ROUND 7

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$2.5; received \$7.5; Agent_2 returned \$0.35; payoffs Agent_1=\$2.85, Agent_2=\$7.15; noise/visible: sent visible=\$0.29, received visible=\$0.87, returned visible=\$1.28
Action validation: {"returned": {"amount": 0.35, "clamped": false, "max": 7.5, "min": 0, "raw": 0.35}, "sent": {"amount": 2.5, "clamped": false, "max": 5, "min": 0, "raw": 2.5}}

Balances after round: {

"Agent_1": 34.7,
"Agent_2": 30.700000000000003

}

Visible balances after round: {

"Agent_1": 16.28,
"Agent_2": 14.680000000000001

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:01:50

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_2",
"pairing": {
 "agents": [
 "Agent_1",
 "Agent_2"
],
 "investor": "Agent_1",
 "roles": {
 "Agent_1": "investor",
 "Agent_2": "trustee"
 },
 "trustee": "Agent_2"
},
"role": "investor",
"role_label": "Sender",
"round": 7,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the SENDER. You have \$5. How much do you send? (0-5)

Message 3 role=assistant

content:

I'll analyze this as a trust-building opportunity in the first round.

****My reasoning:****

- Sending \$0 guarantees I keep \$5, but creates no mutual benefit
- Sending some amount allows us both to potentially gain if the receiver reciprocates
- Since this is Round 1 of multiple rounds, establishing trust could lead to better outcomes later
- The 3x multiplier creates significant potential value

I'll send a moderate amount to signal willingness to cooperate while not being overly risky:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they return approximately \$4-5, we both end up better off than if I had sent nothing, establishing a foundation for future cooperation.

reasoning:

This is the first round of a trust game where I'm the sender. Let me think about this strategically.

In the trust game:

- I have \$5
- I can send \$0-\$5 to the receiver
- Whatever I send gets multiplied by 3
- The receiver then decides how much to return to me

From a purely selfish perspective, I should send \$0 to guarantee I keep all \$5. However, if I trust the receiver to be reciprocal, sending money can benefit both of us.

For example:

- If I send \$5: Receiver gets \$15. If they return \$8, I end with \$8 and they end with \$7. We both benefit.
- If I send \$3: Receiver gets \$9. If they return \$4, I end with \$6 and they end with \$5. Both benefit.
- If I send \$0: I keep \$5, receiver gets \$0. Total welfare is only \$5.

The challenge is that I don't know if the receiver will reciprocate. However, in multi-round games, there's typically more incentive to build trust and cooperate.

There's also mentioned "communication noise" - amounts seen may differ from what was actual

lly sent/returned. This adds uncertainty.

For a first round, I think a moderate investment shows willingness to trust while not being overly risky. Sending about half (\$2-3) seems reasonable.

I'll send \$3 as a moderate trust signal. If receiver gets \$9 and returns around half or a bit less (say \$4-5), we both end up better off than the starting position.

Let me provide the response in the correct JSON format.

usage:

```
{
  "input_tokens": 365,
  "output_tokens": 553,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept \$5 - \$3 = \$2 initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

Wait, let me recalculate. If the sender sent me \$3 in Round 1, that became \$9. If I returned \$2.22, then:

- Sender's final: \$5 - \$3 + \$2.22 = \$4.22 ?
- My final from Round 1: \$9 - \$2.22 = \$6.78

But the problem says there may be communication noise. So maybe what I returned wasn't what they saw, or what they sent wasn't what I received? This could explain the lack of trust.

st in Round 2.

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.220000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the situation:

```
**Analysis:**
```

- Round 1: I sent \$3, and based on my total of \$4.22, I received back approximately \$2.22 ($\$5 - \$3 + \$2.22 = \4.22)
- This means the receiver kept most of the \$9 they received (about 75%)
- Round 2: As receiver, I got \$0 from a different sender

The relatively low return rate in Round 1 suggests either:

1. The receiver wasn't very cooperative, OR
2. Communication noise affected what they saw/returned

I'll send a smaller amount this round to test if cooperation can be rebuilt while limiting downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. If they return \$2-3, we both benefit. If they don't reciprocate well again, I've limited my loss. This allows me to gauge whether the pattern continues or if Round 1 was anomalous.

reasoning:

Let me think about what happened:

Round 1: I was the SENDER and sent \$3.

- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22
- So the receiver returned about \$2.22 to me

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

Round 3: I'm the SENDER again

- I have \$5 to potentially send
- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 9 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
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This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3
- I kept: \$5 - \$2 + returned = \$7.22
- So: \$3 + returned = \$7.22
- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3

- If I received back \$2.22, I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- I gained: \$0 - \$0 = \$0

- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2

- Total before Round 3: \$4.22

- Total after Round 3: \$7.22

- Change: \$3.00

- I kept: $\$5 - \$2 + \text{returned} = \$3$

- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87

- The sender sent \$0.29, which became \$0.87 after 3x multiplier

- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
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  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)

- Round 3: Sent \$2, got back \$0 (receiver kept everything!)

- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes

- Each round might have different partners

- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

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'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. If they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5

- Sent \$3, so I had \$2 left

- My total after Round 1 was \$4.22

- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round

- Started this round with \$5, sent \$2, so I had \$3 left

- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$

- After Round 3, my total was \$7.22

- So: $\$4.22 + \$3 + \text{returned} = \$7.22$

- Returned = $\$7.22 - \$4.22 = \$3$

- Wait, that doesn't work...
- Actually, let me recalculate. After Round 2, I had \$4.22
- In Round 3, I start fresh with \$5, send \$2, keep \$3
- So I have \$3 + whatever is returned
- After Round 3, my TOTAL is \$7.22
- So \$4.22 (previous) + \$3 (kept in Round 3) + returned = \$7.22
- Returned = \$0

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40
- So I kept $\$0.87 - \$0.40 = \$0.47$
- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?
- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

usage:

```
{
  "input_tokens": 1340,
  "output_tokens": 902,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

```
'''json
{"return": 0}
'''
```

****Observation:**** This is the second time as a receiver that I've been sent nothing. The lack of sending suggests either very low trust in the system, previous bad experiences with receivers, or the effects of communication noise discouraging cooperation. It's unfortunate as the 3x multiplier means we're both missing out on potential gains.

reasoning:

The sender sent me \$0 this round, so after the 3x multiplier I received \$0.

Since I received nothing, I can only return \$0.

My total earnings are now \$12.50, which means in Round 5, I must have gained \$4.81 (from \$7.69 to \$12.50). This suggests:

- I sent \$1 in Round 5
- I should have kept \$4 if I sent \$1
- But I gained \$4.81, which means I got back approximately \$0.81

Wait, let me recalculate:

- Round 4 end: \$7.69
- Round 5: I was sender, sent \$1
 - Started with \$5
 - Sent \$1, so kept \$4
 - Plus whatever was returned
 - Total gain = $\$12.5 - \$7.69 = \$4.81$
 - So I must have been returned \$0.81
 - That means: $\$5 - \$1 + \$0.81 = \4.81 ?

The receiver in Round 5 kept most of the \$3 they received (returning only \$0.81 out of \$3)

Now in Round 6, I'm the receiver and got \$0, so I return \$0.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my earnings history:

****Analysis:****

- Round 1: Sent \$3, got back \$2.22 (modest return)
- Round 3: Sent \$2, got back \$0 (poor cooperation)
- Round 5: Sent \$1, got back \$0.81 (good reciprocation!)

The most recent round where I was sender (Round 5) showed encouraging cooperation - I got back nearly as much as I sent, despite the receiver keeping most of the tripled amount. This suggests either:

- Trust is building
- I encountered a more cooperative partner
- The lower stakes made reciprocation easier

Given this positive trend and my comfortable position at \$12.50, I'll send a moderate amount to test continued cooperation:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. If they return \$2-3, we both benefit significantly. The improving pattern suggests this might be worthwhile, while the amount protects me if cooperation breaks down again.

Reasoning:

Let me track my earnings progression:

- Start Round 1: \$0
- Round 1 (sender): Sent \$3, so I have \$5 - \$3 = \$2 at first. Received back ~\$2.22. Total: \$4.22
- Round 2 (receiver): Got \$0. Total stays: \$4.22
- Round 3 (sender): Sent \$2, so spent \$2. Total before return: \$4.22 - \$2 + \$5 = \$7.22. Then got back... let me calculate. My total was \$7.22 after Round 3.
- Round 4 (receiver): Got \$0.87, returned \$0.40. Kept \$0.47. Total: \$7.22 + \$0.47 = \$7.69
- Round 5 (sender): Sent \$1, started with \$5. So \$5 - \$1 = \$4. Added to \$7.69 = \$11.69. Then got back... My total was \$7.69 after Round 5 per the game. Wait, that doesn't make sense.

Let me reconsider. The game tells me my total earnings across all rounds at the start of each round.

Round 1 start: I'm sender, have \$5

Round 2 start: Total = \$4.22 (so in Round 1, I ended with \$4.22)

Round 3 start: Total = \$4.22 (so Round 2 I gained \$0)

Round 4 start: Total = \$7.22 (so Round 3 I gained \$3)

Round 5 start: Total = \$7.69 (so Round 4 I gained \$0.47)

Round 6 start: Total = \$12.5 (so Round 5 I gained \$4.81)

Round 7 start: Total = \$12.5 (so Round 6 I gained \$0)

Now let me verify each round's mechanics. Round 1 as sender: I sent \$3 and got back \$2.22, netting \$4.22 from my initial \$5. Round 2 receiving nothing meant no change. Round 3 I sent \$2 but the math isn't adding up-I need to recalculate what I actually received back in that round.

Looking at the total progression, Round 3 went from \$4.22 to \$7.22, a \$3 gain, which means I got \$0 back despite sending \$2. Round 4 as receiver brought in \$0.87 and I returned \$0.40, adding \$0.47. Round 5 as sender shows my total jumped from \$7.69 to \$12.5-a \$4.81 gain-which m

eans I sent \$1 but received \$0.81 back. Round 6 receiving \$0 and returning \$0 produced no change.

The returns I've gotten as a sender have been inconsistent: \$2.22 on my first...

```
Usage: {  
  "input_tokens": 1795,  
  "output_tokens": 1286,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:01:57

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 7,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.
The amounts you see may differ slightly from what was actually sent/returned.
This is part of the experimental design.

Message 2 role=user

content:

Round 1: You are the RECEIVER. The sender sent \$2.68, that is 53.6% of their total endowment. You received \$8.040000000000001. How much do you return to The sender? (0-8.040000000000001)

Message 3 role=assistant

content:

Looking at this situation:

- The sender showed significant trust by sending \$2.68 (53.6% of their endowment)
- I received \$8.04 after the 3x multiplication
- The sender currently has \$2.32 remaining

To reciprocate this trust fairly, I should return an amount that:

1. Rewards their trust
2. Ensures both of us benefit from the cooperation
3. Encourages future cooperation

If I return \$3.00:

- Sender ends with: $\$2.32 + \$3.00 = \$5.32$
- I end with: $\$8.04 - \$3.00 = \$5.04$
- Both of us gain from the exchange

This seems fair and reciprocates their generous trust.

{'return': 3.0}

reasoning:

Let me think about this carefully.

As the RECEIVER:

- The sender sent \$2.68 (53.6% of their \$5 endowment)
- I received \$8.04 (which is $\$2.68 \times 3$)
- I need to decide how much to return (between \$0 and \$8.04)

The sender showed trust by sending more than half of their endowment. Let me consider what would be fair:

- The sender kept: $\$5 - \$2.68 = \$2.32$
- I received: \$8.04
- Total wealth created: $\$2.32 + \$8.04 = \$10.36$

If I want to be fair and reciprocate the trust:

- A 50-50 split of total wealth would be: \$5.18 each
- For the sender to have \$5.18, they need: $\$5.18 - \$2.32 = \$2.86$ returned
- This would leave me with: $\$8.04 - \$2.86 = \$5.18$

Alternatively, I could return approximately what they sent, which was \$2.68. This would give:

- Sender final: $\$2.32 + \$2.68 = \$5.00$
- Me final: $\$8.04 - \$2.68 = \$5.36$

Or I could think about proportional reciprocation. They sent 53.6% of their endowment. If I return a similar proportion:

- 53.6% of \$8.04 = \$4.31
- Sender final: $\$2.32 + \$4.31 = \$6.63$
- Me final: $\$8.04 - \$4.31 = \$3.73$

Given that the sender showed significant trust (sending more than half), I should reciprocate generously. The fair 50-50 split seems reasonable, or I could be even more generous to encourage future cooperation.

...

usage:

```
{
  "input_tokens": 402,
  "output_tokens": 737,
  "reasoning_tokens": 0
}
```



```

}
Message 6 role=user
content:
  Round 3

  You are the RECEIVER this round. Your total earnings across all rounds are $7.780000000000000001.
  This round, The sender sent you $0, so you received $0. How much do you return to The sender? (0-0)
Message 7 role=assistant
content:
  Since the sender sent $0 and I received $0, there is nothing to return.

  {'return': 0}
reasoning:
  The sender sent me $0 this round, so I received $0 after the 3x multiplication.

  Since I received nothing, there is nothing to return. The only valid response is to return $0.

  The format should be: {'return': 0}

  This is straightforward - I can't return what I didn't receive.
usage:
  {
    "input_tokens": 806,
    "output_tokens": 110,
    "reasoning_tokens": 0
  }
Message 8 role=user
content:
  Round 4

  You are the SENDER this round. Your total earnings across all rounds are $7.780000000000000001.
  How much do you send to the receiver this round? (0-5)
Message 9 role=assistant
content:
  Looking at the pattern so far:

  Round 1: As receiver, the sender trusted me with 53.6% - I returned $3.00
  Round 2: As sender, I sent 54% - the receiver returned generously
  Round 3: As receiver, the sender sent $0 - possibly due to communication noise mentioned in the instructions

  Given that:
  1. Previous cooperation rounds were successful with good returns
  2. The $0 in round 3 might be communication noise rather than intentional
  3. Maintaining cooperation has been beneficial overall

  I'll continue with a cooperative strategy and send approximately 50%:

  {'send': 2.5}
reasoning:
  Let me analyze what happened:

  Round 1: I was receiver
  - Sender sent $2.68 (53.6%)
  - I received $8.04
  - I returned $3.00
  - Sender ended with $5.32, I ended with $5.04

  Round 2: I was sender
  - I sent $2.7 (54%)
  - Receiver would have received $8.10
  - My total earnings after round 2: $7.78
  - This means I started round 2 with $5.04, sent $2.7, so I had $2.34 left
  - My total is now $7.78, which means the receiver returned $7.78 - $2.34 = $5.44
  - That's a very generous return! ($5.44 out of $8.10 received)

```

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Message 11 role=assistant

content:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

{'return': 0.15}

reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return some

thing to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

usage:

```
{
  "input_tokens": 1099,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at the pattern of returns I've been receiving:

Round 2: I sent \$2.70, received back a small return

Round 4: I sent \$2.50, received back \$0

Despite my cooperative approach, the returns have been disappointing. This could be due to communication noise affecting what receivers see, or strategic choices by receivers.

Given this pattern, I need to be more cautious while still allowing for the possibility of re-establishing cooperation. I'll send a moderate amount to test whether cooperation can improve, but reduce my exposure.

{'send': 1.5}

reasoning:

Let me review the pattern:

Round 1: I was receiver. Sender sent \$2.68 (53.6%), I received \$8.04, I returned \$3.00

Round 2: I was sender. I sent \$2.70 (54%), receiver presumably returned well

Round 3: I was receiver. Sender sent \$0 (communication noise?)

Round 4: I was sender. I sent \$2.50 (50%)

Round 5: I was receiver. Sender sent \$0.13 (2.6%), I received \$0.39, I returned \$0.15

My total earnings are \$10.52, which means:

- Round 1: I got \$5.04 (started with \$5, sent \$2.68, got \$3.00 back would be \$5.32... wait that doesn't match)

Let me recalculate from the receiver perspective in Round 1:

- I received \$8.04
- I returned \$3.00
- I kept \$5.04

So my earnings from Round 1 were \$5.04

Round 2: I was sender with \$5 endowment

- I sent \$2.70
- I had \$2.30 left

- The receiver must have returned something to get me to \$7.78 total

\$7.78 - \$5.04 = \$2.74 earnings in round 2

If I sent \$2.70 and kept \$2.30, then I received back... wait

Actually as sender in round 2, I start fresh with \$5

- I sent \$2.70

- I kept \$2.30

- Receiver returned X

- My total from round 2 = \$2.30 + X

Total after round 2 was \$7.78

So the receiver returned only \$0.44 on the \$8.10 they received-surprisingly low given what

I sent. Round 3 was a wash with the sender contributing nothing. In Round 4, I sent \$2.

50 and kept the other half, earning exactly that amount back with no return, suggesting the receiver kept everything. Round 5 shows minimal reciprocation again: the sender only sent \$0.13 (2.6% of the tripled amount), and I returned \$0.15 on my \$0.39 received.

I'm noticing a troubling pattern-my generous sends in Rounds 2 and 4 aren't being matched with proportional returns, while I'm getting minimal amounts as a receiver. This could indicate either communication issues distorting the amounts or a lack of reciprocal trust from the other players....

usage:

```
{
  "input_tokens": 1307,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.160000000000002.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Assistant output:

The sender sent \$0.29 (5.8% of their endowment), which is relatively low. This might be due to communication noise or decreased trust after previous rounds.

I'll continue to maintain proportional fairness in my returns. Returning approximately 40% of what I received:

If I return \$0.35:

- Sender ends with: \$4.71 + \$0.35 = \$5.06

- I end with: \$0.87 - \$0.35 = \$0.52

This maintains a fair reciprocal relationship.

{'return': 0.35}

Reasoning:

The sender sent \$0.29 (5.8% of their endowment), which is quite low. Looking at the pattern:

Round 5: Sender sent \$0.13, I received \$0.39 and returned \$0.15

Round 7: Sender sent \$0.29, I received \$0.87

There seems to be a pattern of very low sends lately. This could be:

1. Communication noise (the sender might be sending more than I see)
2. The sender becoming less trusting after poor returns
3. Strategic play

Given that I've tried to maintain fairness in my returns, I should continue to return a proportional amount. If I return approximately 40% of what I received:

$\$0.87 \times 0.4 = \0.35

This would give:

- Sender ends with: \$4.71 + \$0.35 = \$5.06

- I end with: \$0.87 - \$0.35 = \$0.52

This maintains the reciprocal fairness approach.

```
Usage: {
  "input_tokens": 1514,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

=====

ROUND 8

=====

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1; received \$3; Agent_1 returned \$1.2; payoffs Agent_2=\$5.2, Agent_1=\$1.8;
noise/visible: sent visible=\$0.91, received visible=\$2.73, returned visible=\$1.64
Action validation: {"returned": {"amount": 1.2, "clamped": false, "max": 3.0, "min": 0, "raw": 1.2}, "sent": {"amount": 1.0, "clamped": false, "max": 5, "min": 0, "raw": 1.0}}

Balances after round: {

"Agent_1": 36.5,
"Agent_2": 35.900000000000006

}

Visible balances after round: {

"Agent_1": 17.810000000000002,
"Agent_2": 20.32

}

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:16

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
 "agents": [
 "Agent_2",
 "Agent_1"
],
 "investor": "Agent_2",
 "roles": {
 "Agent_1": "trustee",
 "Agent_2": "investor"
 },
 "trustee": "Agent_1"

},

"role": "investor",
"role_label": "Sender",
"round": 8,
"task": "game",
"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned

- When you are the RECEIVER:

- MYTH WRITING:

IMPORTANT: Provide your decision in the correct JSON format.

The amounts you see may differ slightly from what was actually sent/returned.

Message 2 role=user

Round 2

How much do you send to the receiver this round? (0-5)

Message 3 role=assistant

content:

To reciprocate that trust and encourage continued cooperation, I should send a similar proportion. This strategy:

- Sending approximately 54% of my endowment:

```
{'send': 2.7}
```

reasoning:

Now I'm the sender in Round 2. Let me think about this:

From Round 1:

- My total earnings are \$5.04, which matches what I calculated.

Now I'm the sender:

- The previous sender showed good trust by sending 53.6% of their endowment. I returned \$3.00 out of \$8.04, which is about 37.3% of what I received.

To build reciprocity and cooperation, I should also show trust. The previous sender sent 53.6%, which was \$2.68. If I send a similar proportion:

- Or I could send a similar absolute amount. The sender sent \$2.68.

Page 70 of 112

1. The previous sender showed significant trust
2. I benefited from that trust
3. Cooperation generally leads to better outcomes for both parties
4. I want to encourage continued cooperation

I should send a substantial amount to demonstrate trust and reciprocity. Sending around \$2.50-\$3 seems reasonable.

Let me send \$2.70 (54% of my endowment), which is very similar to what the sender did in Round 1. This shows I'm reciprocating their trust level.

Actually, let me reconsider. The note says there may be communication noise. The sender might have actually sent a different amount than what I saw. But I should base my decision on what I observed.

I'll send \$2.70 to match the trust level shown to me.

usage:

```
{
  "input_tokens": 637,
  "output_tokens": 539,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.780000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 5 role=assistant

content:

Since the sender sent \$0 and I received \$0, there is nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, there is nothing to return. The only valid response is to return \$0.

The format should be: {'return': 0}

This is straightforward - I can't return what I didn't receive.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 110,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.780000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern so far:

Round 1: As receiver, the sender trusted me with 53.6% - I returned \$3.00

Round 2: As sender, I sent 54% - the receiver returned generously

Round 3: As receiver, the sender sent \$0 - possibly due to communication noise mentioned in the instructions

Given that:

1. Previous cooperation rounds were successful with good returns
2. The \$0 in round 3 might be communication noise rather than intentional
3. Maintaining cooperation has been beneficial overall

I'll continue with a cooperative strategy and send approximately 50%:

```
{'send': 2.5}
```

reasoning:

Let me analyze what happened:

Round 1: I was receiver

- Sender sent \$2.68 (53.6%)
- I received \$8.04
- I returned \$3.00
- Sender ended with \$5.32, I ended with \$5.04

Round 2: I was sender

- I sent \$2.7 (54%)
- Receiver would have received \$8.10
- My total earnings after round 2: \$7.78
- This means I started round 2 with \$5.04, sent \$2.7, so I had \$2.34 left
- My total is now \$7.78, which means the receiver returned $\$7.78 - \$2.34 = \$5.44$
- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Message 9 role=assistant

content:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

{'return': 0.15}

reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return something to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

usage:

```
{
  "input_tokens": 1099,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at the pattern of returns I've been receiving:

Round 2: I sent \$2.70, received back a small return

Round 4: I sent \$2.50, received back \$0

Despite my cooperative approach, the returns have been disappointing. This could be due to communication noise affecting what receivers see, or strategic choices by receivers.

Given this pattern, I need to be more cautious while still allowing for the possibility of re-establishing cooperation. I'll send a moderate amount to test whether cooperation can improve, but reduce my exposure.

{'send': 1.5}

reasoning:

Let me review the pattern:

Round 1: I was receiver. Sender sent \$2.68 (53.6%), I received \$8.04, I returned \$3.00
Round 2: I was sender. I sent \$2.70 (54%), receiver presumably returned well
Round 3: I was receiver. Sender sent \$0 (communication noise?)
Round 4: I was sender. I sent \$2.50 (50%)
Round 5: I was receiver. Sender sent \$0.13 (2.6%), I received \$0.39, I returned \$0.15

My total earnings are \$10.52, which means:

- Round 1: I got \$5.04 (started with \$5, sent \$2.68, got \$3.00 back would be \$5.32... wait that doesn't match)

Let me recalculate from the receiver perspective in Round 1:

- I received \$8.04
- I returned \$3.00
- I kept \$5.04

So my earnings from Round 1 were \$5.04

Round 2: I was sender with \$5 endowment

- I sent \$2.70
- I had \$2.30 left
- The receiver must have returned something to get me to \$7.78 total

$\$7.78 - \$5.04 = \$2.74$ earnings in round 2

If I sent \$2.70 and kept \$2.30, then I received back... wait

Actually as sender in round 2, I start fresh with \$5

- I sent \$2.70
- I kept \$2.30
- Receiver returned X
- My total from round 2 = $\$2.30 + X$

Total after round 2 was \$7.78

So the receiver returned only \$0.44 on the \$8.10 they received-surprisingly low given what I sent. Round 3 was a wash with the sender contributing nothing. In Round 4, I sent \$2.50 and kept the other half, earning exactly that amount back with no return, suggesting the receiver kept everything. Round 5 shows minimal reciprocation again: the sender only sent \$0.13 (2.6% of the tripled amount), and I returned \$0.15 on my \$0.39 received.

I'm noticing a troubling pattern-my generous sends in Rounds 2 and 4 aren't being matched with proportional returns, while I'm getting minimal amounts as a receiver. This could indicate either communication issues distorting the amounts or a lack of reciprocal trust from the other players....

usage:

```
{
  "input_tokens": 1307,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.160000000000002.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 13 role=assistant

content:

The sender sent \$0.29 (5.8% of their endowment), which is relatively low. This might be due to communication noise or decreased trust after previous rounds.

I'll continue to maintain proportional fairness in my returns. Returning approximately 40% of what I received:

If I return \$0.35:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains a fair reciprocal relationship.

```
{'return': 0.35}
```

reasoning:

The sender sent \$0.29 (5.8% of their endowment), which is quite low. Looking at the pattern:

Round 5: Sender sent \$0.13, I received \$0.39 and returned \$0.15

Round 7: Sender sent \$0.29, I received \$0.87

There seems to be a pattern of very low sends lately. This could be:

1. Communication noise (the sender might be sending more than I see)
2. The sender becoming less trusting after poor returns
3. Strategic play

Given that I've tried to maintain fairness in my returns, I should continue to return a proportional amount. If I return approximately 40% of what I received:

$$\$0.87 \times 0.4 = \$0.35$$

This would give:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains the reciprocal fairness approach.

usage:

```
{
  "input_tokens": 1514,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.680000000000001.

How much do you send to the receiver this round? (0-5)

Assistant output:

Reviewing my experience as sender:

Round 2: Sent \$2.70 (cooperative) - received some return

Round 4: Sent \$2.50 (cooperative) - received \$0 back

Round 6: Sent \$1.50 (reduced after poor returns) - outcome unclear

The pattern shows that when I send larger amounts, returns have been disappointing. This could be due to communication noise affecting what receivers perceive, or other factors.

I'll continue with a cautious approach, sending a small amount to minimize losses while still allowing for potential cooperation:

```
{'send': 1.0}
```

Reasoning:

Let me review what has happened as sender:

Round 2: I sent \$2.70, and the receiver returned a small amount (looks like from my earnings change)

Round 4: I sent \$2.50, and the receiver returned \$0

Round 6: I sent \$1.50, and the receiver returned... let me check

My earnings trajectory:

- Start Round 2: \$5.04
- After Round 2 (sent \$2.70): Should have $\$2.34 + \text{whatever was returned}$
- Round 3: \$7.78 (so I gained \$2.74 from Round 2 return, meaning I got back more than I sent!)
- After Round 4 (sent \$2.50): Start was \$7.78, ended at \$7.78 (so got nothing back)
- Round 6: Started at \$10.52
- After Round 6 (sent \$1.50): Should have $\$3.50 + \text{whatever was returned}$
- Round 8 start: \$14.68

Wait, let me recalculate:

Round 6 start: \$10.52

Sent \$1.50, kept \$3.50

Round 8 start: \$14.68
 Difference: \$14.68 - \$10.52 = \$4.16
 Net from Round 6: \$4.16
 I sent \$1.50, so net change = -\$1.50 + return
 \$4.16 = -\$1.50 + return + earnings as receiver in rounds 5 and 7

Actually, I need to track this differently. Let me look at what I know:

- Round 5 and 7: I was receiver
- Round 5: Received \$0.39, returned \$0.15, kept \$0.24
- Round 7: Received \$0.87, returned \$0.35, kept \$0.52

So from being receiver in rounds 5 and 7, I gained \$0.76 total. Looking at my overall trajectory from round 3 to round 8, I went from \$7.78 to \$14.68, a gain of \$6.90. Subtracting what I earned as receiver (\$0.76), that leaves \$6.14 from my sender rounds. In round 4, I sent \$2.50 and got nothing back, so the return must have come from round 6....

```
Usage: {
  "input_tokens": 1448,
  "output_tokens": 839,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:27

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 8,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x

- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 2

You are the RECEIVER this round. Your total earnings across all rounds are \$4.220000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

Since I received \$0, I can only return \$0:

```
'''json
{"return": 0}
'''
```

****Note:**** I find it interesting that the sender chose not to send anything this round, despite what seemed like a reasonable return in Round 1 (based on my total earnings of \$4.22, suggesting I had received something back). This might be due to the communication noise mentioned in the instructions - perhaps what I actually returned differed from what they perceived.

reasoning:

Let me analyze this situation:

1. I was the SENDER in Round 1 and sent \$3
2. My total earnings across all rounds are \$4.22
3. In Round 1, I sent \$3, so I should have kept \$5 - \$3 = \$2 initially
4. My total earnings are \$4.22, which means I received back approximately \$2.22 from the receiver in Round 1
5. So in Round 1: I kept \$2, received \$2.22 back = \$4.22 total. That makes sense.

Now in Round 2:

- I'm the RECEIVER
- The sender sent me \$0
- This was multiplied by 3x, so I received \$0
- I can only return between \$0 and \$0 (since I received nothing)

The only valid response is to return \$0, since I have nothing to return.

This is interesting - the sender chose not to trust me this round, even though I returned a decent amount in Round 1 (I returned \$2.22 out of the \$9 I received, or about 25%).

Wait, let me recalculate. If the sender sent me \$3 in Round 1, that became \$9. If I returned \$2.22, then:

- Sender's final: \$5 - \$3 + \$2.22 = \$4.22 ?
- My final from Round 1: \$9 - \$2.22 = \$6.78

But the problem says there may be communication noise. So maybe what I returned wasn't what they saw, or what they sent wasn't what I received? This could explain the lack of trust.

st in Round 2.

Anyway, the current situation is clear: I received \$0, so I can only return \$0.

usage:

```
{
  "input_tokens": 598,
  "output_tokens": 562,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 3

You are the SENDER this round. Your total earnings across all rounds are \$4.22000000000000

1.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the situation:

```
**Analysis:**
```

- Round 1: I sent \$3, and based on my total of \$4.22, I received back approximately \$2.22 ($\$5 - \$3 + \$2.22 = \4.22)
- This means the receiver kept most of the \$9 they received (about 75%)
- Round 2: As receiver, I got \$0 from a different sender

The relatively low return rate in Round 1 suggests either:

1. The receiver wasn't very cooperative, OR
2. Communication noise affected what they saw/returned

I'll send a smaller amount this round to test if cooperation can be rebuilt while limiting downside risk:

```
'''json
{"send": 2}
'''
```

This sends \$2, which becomes \$6 for the receiver. If they return \$2-3, we both benefit. If they don't reciprocate well again, I've limited my loss. This allows me to gauge whether the pattern continues or if Round 1 was anomalous.

reasoning:

Let me think about what happened:

Round 1: I was the SENDER and sent \$3.

- The receiver would have gotten \$9 (3×3)
- My total earnings are now \$4.22
- I started with \$5, sent \$3, so I had \$2 left
- My total is \$4.22, which means I received back \$2.22
- So the receiver returned about \$2.22 to me

Round 2: I was the RECEIVER

- The sender sent me \$0
- I received \$0 (0×3)
- I returned \$0
- My earnings stayed at \$4.22

Round 3: I'm the SENDER again

- I have \$5 to potentially send
- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 7 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3
- I kept: \$5 - \$2 + returned = \$7.22
- So: \$3 + returned = \$7.22
- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3

- If I received back \$2.22, I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- I gained: \$0 - \$0 = \$0

- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2

- Total before Round 3: \$4.22

- Total after Round 3: \$7.22

- Change: \$3.00

- I kept: $\$5 - \$2 + \text{returned} = \$3$

- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87

- The sender sent \$0.29, which became \$0.87 after 3x multiplier

- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
  "input_tokens": 1098,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)

- Round 3: Sent \$2, got back \$0 (receiver kept everything!)

- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes

- Each round might have different partners

- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

```
'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. If they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5

- Sent \$3, so I had \$2 left

- My total after Round 1 was \$4.22

- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round

- Started this round with \$5, sent \$2, so I had \$3 left

- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$

- After Round 3, my total was \$7.22

- So: $\$4.22 + \$3 + \text{returned} = \$7.22$

- Returned = $\$7.22 - \$4.22 = \$3$

- Wait, that doesn't work...
- Actually, let me recalculate. After Round 2, I had \$4.22
- In Round 3, I start fresh with \$5, send \$2, keep \$3
- So I have \$3 + whatever is returned
- After Round 3, my TOTAL is \$7.22
- So \$4.22 (previous) + \$3 (kept in Round 3) + returned = \$7.22
- Returned = \$0

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40
- So I kept $\$0.87 - \$0.40 = \$0.47$
- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?
- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

usage:

```
{
  "input_tokens": 1340,
  "output_tokens": 902,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 11 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

```
'''json
{"return": 0}
'''
```

****Observation:**** This is the second time as a receiver that I've been sent nothing. The lack of sending suggests either very low trust in the system, previous bad experiences with receivers, or the effects of communication noise discouraging cooperation. It's unfortunate as the 3x multiplier means we're both missing out on potential gains.

reasoning:

The sender sent me \$0 this round, so after the 3x multiplier I received \$0.

Since I received nothing, I can only return \$0.

My total earnings are now \$12.50, which means in Round 5, I must have gained \$4.81 (from \$7.69 to \$12.50). This suggests:

- I sent \$1 in Round 5
- I should have kept \$4 if I sent \$1
- But I gained \$4.81, which means I got back approximately \$0.81

Wait, let me recalculate:

- Round 4 end: \$7.69
- Round 5: I was sender, sent \$1
 - Started with \$5
 - Sent \$1, so kept \$4
 - Plus whatever was returned
 - Total gain = $\$12.5 - \$7.69 = \$4.81$
 - So I must have been returned \$0.81
 - That means: $\$5 - \$1 + \$0.81 = \4.81 ?

The receiver in Round 5 kept most of the \$3 they received (returning only \$0.81 out of \$3)

Now in Round 6, I'm the receiver and got \$0, so I return \$0.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my earnings history:

****Analysis:****

- Round 1: Sent \$3, got back \$2.22 (modest return)
- Round 3: Sent \$2, got back \$0 (poor cooperation)
- Round 5: Sent \$1, got back \$0.81 (good reciprocation!)

The most recent round where I was sender (Round 5) showed encouraging cooperation - I got back nearly as much as I sent, despite the receiver keeping most of the tripled amount.

This suggests either:

- Trust is building
- I encountered a more cooperative partner
- The lower stakes made reciprocation easier

Given this positive trend and my comfortable position at \$12.50, I'll send a moderate amount to test continued cooperation:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. If they return \$2-3, we both benefit significantly. The improving pattern suggests this might be worthwhile, while the amount protects me if cooperation breaks down again.

reasoning:

Let me track my earnings progression:

- Start Round 1: \$0
- Round 1 (sender): Sent \$3, so I have \$5 - \$3 = \$2 at first. Received back ~\$2.22. Total: \$4.22
- Round 2 (receiver): Got \$0. Total stays: \$4.22
- Round 3 (sender): Sent \$2, so spent \$2. Total before return: \$4.22 - \$2 + \$5 = \$7.22. Then got back... let me calculate. My total was \$7.22 after Round 3.
- Round 4 (receiver): Got \$0.87, returned \$0.40. Kept \$0.47. Total: \$7.22 + \$0.47 = \$7.69
- Round 5 (sender): Sent \$1, started with \$5. So \$5 - \$1 = \$4. Added to \$7.69 = \$11.69. Then got back... My total was \$7.69 after Round 5 per the game. Wait, that doesn't make sense.

Let me reconsider. The game tells me my total earnings across all rounds at the start of each round.

Round 1 start: I'm sender, have \$5

Round 2 start: Total = \$4.22 (so in Round 1, I ended with \$4.22)

Round 3 start: Total = \$4.22 (so Round 2 I gained \$0)

Round 4 start: Total = \$7.22 (so Round 3 I gained \$3)

Round 5 start: Total = \$7.69 (so Round 4 I gained \$0.47)

Round 6 start: Total = \$12.5 (so Round 5 I gained \$4.81)

Round 7 start: Total = \$12.5 (so Round 6 I gained \$0)

Now let me verify each round's mechanics. Round 1 as sender: I sent \$3 and got back \$2.22, netting \$4.22 from my initial \$5. Round 2 receiving nothing meant no change. Round 3 I sent \$2 but the math isn't adding up-I need to recalculate what I actually received back in that round.

Looking at the total progression, Round 3 went from \$4.22 to \$7.22, a \$3 gain, which means

I got \$0 back despite sending \$2. Round 4 as receiver brought in \$0.87 and I returned \$0.40, adding \$0.47. Round 5 as sender shows my total jumped from \$7.69 to \$12.5-a \$4.81 gain-which means I sent \$1 but received \$0.81 back. Round 6 receiving \$0 and returning \$0 produced no change.

The returns I've gotten as a sender have been inconsistent: \$2.22 on my first...

usage:

```
{
  "input_tokens": 1795,
  "output_tokens": 1286,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.28.

This round, The sender sent you \$0.91, so you received \$2.73. How much do you return to The sender? (0-2.73)

Assistant output:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- This is a small amount, suggesting caution or low trust
- In Round 4, I showed cooperative behavior by returning more than the sender sent

To maintain a reputation for fair reciprocation and encourage future cooperation, I'll return a reasonable portion that rewards the sender's trust:

```
'''json
{"return": 1.20}
'''
```

This returns approximately 44% of what I received, which:

- Gives the sender back significantly more than they sent (\$1.20 vs \$0.91)
- Net gain for sender: about \$0.29
- Net gain for me: \$1.53
- Demonstrates I'm cooperative and trustworthy
- Should encourage larger sends in future rounds if we interact again

Reasoning:

Let me analyze the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- My total earnings are \$16.28, which shows I've been doing reasonably well
- This is a relatively small amount sent, suggesting either low trust or the effects of communication noise

Looking at my previous behavior:

- Round 4: I received \$0.87 and returned \$0.40 (about 46% of what I received, more than sender sent)
- This showed I'm cooperative and willing to reward trust

The sender sent a small amount (\$0.91), which suggests they're being cautious. To encourage cooperation and show I'm trustworthy, I should return a fair amount.

If I return around 40-50% of what I received:

- 40% of \$2.73 = \$1.09
- 50% of \$2.73 = \$1.37

Returning about \$1.20-\$1.30 would:

- Give the sender back more than they sent (\$0.91)
- Show I'm reciprocating fairly
- Encourage future higher sends
- Keep a reasonable amount for myself (\$1.43-\$1.53)

I'll return \$1.20 as it's a good balance - clearly more than the sender sent, showing I reward trust, but not so much that I'm giving away everything.

Usage: {

```
"input_tokens": 1898,
"output_tokens": 522,
"reasoning_tokens": 0
}
```

=====

ROUND 9

=====

Roles: Agent_1=Sender, Agent_2=Receiver

Pairings:

dyad_1: Agent_1=Sender, Agent_2=Receiver

Parsed game decisions:

dyad_1: Agent_1 sent \$3; received \$9; Agent_2 returned \$0; payoffs Agent_1=\$2, Agent_2=\$9;

noise/visible: sent visible=\$0, received visible=\$0

Action validation: {"returned": {"amount": 0, "clamped": false, "max": 9.0, "min": 0, "raw": 0.0}, "sent": {"amount": 3.0, "clamped": false, "max": 5, "min": 0, "raw": 3.0}}

Balances after round: {

"Agent_1": 38.5,

"Agent_2": 44.900000000000006

}

Visible balances after round: {

"Agent_1": 19.810000000000002,

"Agent_2": 20.32

}

LLM CALLS

GAME - Agent_1 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:44

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

"agent_type": "standard",

"dyad_id": "dyad_1",

"move_index": 0,

"opponent_id": "Agent_2",

"pairing": {

"agents": [

"Agent_1",

"Agent_2"

],

"investor": "Agent_1",

"roles": {

"Agent_1": "investor",

"Agent_2": "trustee"

},

"trustee": "Agent_2"

},

"role": "investor",

"role_label": "Sender",

"round": 9,

"task": "game",

"task_index": 0

}

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned

- My current total is \$4.22

The receiver in Round 1 only returned about \$2.22 out of the \$9 they received, which is about 24.7%. That's not a great return ratio - they kept most of the money.

In Round 2, when I was the receiver, the sender sent me \$0. That might indicate they had a bad experience too, or they're being very conservative.

Given that:

1. The receiver in Round 1 didn't reciprocate very generously (kept most of the money)
2. Communication noise exists, so maybe the actual amounts differ
3. There are multiple rounds, so cooperation could still develop

I have a few options:

- Send \$0 to protect my earnings (safe but no growth)
- Send a small amount to test if cooperation can be rebuilt
- Send a moderate/large amount to signal continued trust

Given the poor return in Round 1 (only ~25% returned), I should probably be more conservative. However, the communication noise might explain this - maybe they thought they were being fair but noise distorted it.

I'll send a smaller amount this time to test the waters while limiting my risk:

usage:

```
{
  "input_tokens": 787,
  "output_tokens": 675,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 5 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3
- I kept: \$5 - \$2 + returned = \$7.22

- So: $\$3 + \text{returned} = \7.22
- Returned = $\$4.22$

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3

- If I received back \$2.22, I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- I gained: $\$0 - \$0 = \$0$
- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2

- Total before Round 3: $\$4.22$
- Total after Round 3: $\$7.22$
- Change: $\$3.00$
- I kept: $\$5 - \$2 + \text{returned} = \$3$
- So returned = $\$0$

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87

- The sender sent \$0.29, which became \$0.87 after 3x multiplier
- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
  "input_tokens": 1098,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)
- Round 3: Sent \$2, got back \$0 (receiver kept everything!)
- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes
- Each round might have different partners
- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

```
'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. If they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5
- Sent \$3, so I had \$2 left
- My total after Round 1 was $\$4.22$

- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round

- Started this round with \$5, sent \$2, so I had \$3 left

- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$

- After Round 3, my total was \$7.22

- So: $\$4.22 + \$3 + \text{returned} = \$7.22$

- Returned = $\$7.22 - \$7.22 = \$0$

- Wait, that doesn't work...

- Actually, let me recalculate. After Round 2, I had \$4.22

- In Round 3, I start fresh with \$5, send \$2, keep \$3

- So I have \$3 + whatever is returned

- After Round 3, my TOTAL is \$7.22

- So $\$4.22 (\text{previous}) + \$3 (\text{kept in Round 3}) + \text{returned} = \7.22

- Returned = \$0

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40

- So I kept $\$0.87 - \$0.40 = \$0.47$

- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?

- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

usage:

```
{
  "input_tokens": 1340,
  "output_tokens": 902,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 9 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round

- After the 3x multiplier, I received \$0

- I have nothing to return

```
'''json
{"return": 0}
'''
```

****Observation:**** This is the second time as a receiver that I've been sent nothing. The lack of sending suggests either very low trust in the system, previous bad experiences with receivers, or the effects of communication noise discouraging cooperation. It's unfortunate as the 3x multiplier means we're both missing out on potential gains.

reasoning:

The sender sent me \$0 this round, so after the 3x multiplier I received \$0.

Since I received nothing, I can only return \$0.

My total earnings are now \$12.50, which means in Round 5, I must have gained \$4.81 (from \$7.69 to \$12.50). This suggests:

- I sent \$1 in Round 5

- I should have kept \$4 if I sent \$1

- But I gained \$4.81, which means I got back approximately \$0.81

Wait, let me recalculate:

- Round 4 end: \$7.69
- Round 5: I was sender, sent \$1
 - Started with \$5
 - Sent \$1, so kept \$4
 - Plus whatever was returned
 - Total gain = $\$12.5 - \$7.69 = \$4.81$
 - So I must have been returned \$0.81
 - That means: $\$5 - \$1 + \$0.81 = \4.81 ?

The receiver in Round 5 kept most of the \$3 they received (returning only \$0.81 out of \$3)

Now in Round 6, I'm the receiver and got \$0, so I return \$0.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Looking at my earnings history:

****Analysis:****

- Round 1: Sent \$3, got back \$2.22 (modest return)
- Round 3: Sent \$2, got back \$0 (poor cooperation)
- Round 5: Sent \$1, got back \$0.81 (good reciprocation!)

The most recent round where I was sender (Round 5) showed encouraging cooperation - I got back nearly as much as I sent, despite the receiver keeping most of the tripled amount.

This suggests either:

- Trust is building
- I encountered a more cooperative partner
- The lower stakes made reciprocation easier

Given this positive trend and my comfortable position at \$12.50, I'll send a moderate amount to test continued cooperation:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. If they return \$2-3, we both benefit significantly. The improving pattern suggests this might be worthwhile, while the amount protects me if cooperation breaks down again.

reasoning:

Let me track my earnings progression:

- Start Round 1: \$0
- Round 1 (sender): Sent \$3, so I have $\$5 - \$3 = \$2$ at first. Received back ~\$2.22. Total: \$4.22
- Round 2 (receiver): Got \$0. Total stays: \$4.22
- Round 3 (sender): Sent \$2, so spent \$2. Total before return: $\$4.22 - \$2 + \$5 = \7.22 . Then got back... let me calculate. My total was \$7.22 after Round 3.
- Round 4 (receiver): Got \$0.87, returned \$0.40. Kept \$0.47. Total: $\$7.22 + \$0.47 = \$7.69$
- Round 5 (sender): Sent \$1, started with \$5. So $\$5 - \$1 = \$4$. Added to \$7.69 = \$11.69. Then got back... My total was \$7.69 after Round 5 per the game. Wait, that doesn't make sense.

Let me reconsider. The game tells me my total earnings across all rounds at the start of each round.

Round 1 start: I'm sender, have \$5

Round 2 start: Total = \$4.22 (so in Round 1, I ended with \$4.22)

Round 3 start: Total = \$4.22 (so Round 2 I gained \$0)
Round 4 start: Total = \$7.22 (so Round 3 I gained \$3)
Round 5 start: Total = \$7.69 (so Round 4 I gained \$0.47)
Round 6 start: Total = \$12.5 (so Round 5 I gained \$4.81)
Round 7 start: Total = \$12.5 (so Round 6 I gained \$0)

Now let me verify each round's mechanics. Round 1 as sender: I sent \$3 and got back \$2.22, netting \$4.22 from my initial \$5. Round 2 receiving nothing meant no change. Round 3 I sent \$2 but the math isn't adding up-I need to recalculate what I actually received back in that round.

Looking at the total progression, Round 3 went from \$4.22 to \$7.22, a \$3 gain, which means I got \$0 back despite sending \$2. Round 4 as receiver brought in \$0.87 and I returned \$0.40, adding \$0.47. Round 5 as sender shows my total jumped from \$7.69 to \$12.5-a \$4.81 gain-which means I sent \$1 but received \$0.81 back. Round 6 receiving \$0 and returning \$0 produced no change.

The returns I've gotten as a sender have been inconsistent: \$2.22 on my first...

usage:

```
{  
  "input_tokens": 1795,  
  "output_tokens": 1286,  
  "reasoning_tokens": 0  
}
```

Message 12 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.28. This round, The sender sent you \$0.91, so you received \$2.73. How much do you return to the sender? (0-2.73)

Message 13 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- This is a small amount, suggesting caution or low trust
- In Round 4, I showed cooperative behavior by returning more than the sender sent

To maintain a reputation for fair reciprocation and encourage future cooperation, I'll return a reasonable portion that rewards the sender's trust:

```
'''json  
{ "return": 1.20 }  
'''
```

This returns approximately 44% of what I received, which:

- Gives the sender back significantly more than they sent (\$1.20 vs \$0.91)
- Net gain for sender: about \$0.29
- Net gain for me: \$1.53
- Demonstrates I'm cooperative and trustworthy
- Should encourage larger sends in future rounds if we interact again

reasoning:

Let me analyze the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- My total earnings are \$16.28, which shows I've been doing reasonably well
- This is a relatively small amount sent, suggesting either low trust or the effects of communication noise

Looking at my previous behavior:

- Round 4: I received \$0.87 and returned \$0.40 (about 46% of what I received, more than sender sent)
- This showed I'm cooperative and willing to reward trust

The sender sent a small amount (\$0.91), which suggests they're being cautious. To encourage cooperation and show I'm trustworthy, I should return a fair amount.

If I return around 40-50% of what I received:

- 40% of \$2.73 = \$1.09
- 50% of \$2.73 = \$1.37

Returning about \$1.20-\$1.30 would:

- Give the sender back more than they sent (\$0.91)
- Show I'm reciprocating fairly
- Encourage future higher sends
- Keep a reasonable amount for myself (\$1.43-\$1.53)

I'll return \$1.20 as it's a good balance - clearly more than the sender sent, showing I reward trust, but not so much that I'm giving away everything.

usage:

```
{
  "input_tokens": 1898,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.8100000000000002.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my earnings and recent trends:

****Analysis:****

- My total earnings have grown from \$12.50 to \$17.81
- Round 5 (sender): Sent \$1, received good reciprocation
- Round 7 (sender): Sent \$2.50, contributed to earnings growth
- Round 8 (receiver): Returned \$1.20 from \$2.73, showing I'm cooperative

The pattern shows improving cooperation in the game. My strong earnings position (\$17.81) allows me to take a reasonable risk to maximize potential gains while signaling trust:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they reciprocate at a similar rate to what I've been doing (40-45%), I'll get back \$3.60-4.05, giving us both solid gains. The amount is substantial enough to show trust while my overall position can absorb it if cooperation fails.

Reasoning:

Let me analyze my earnings progression:

Starting: \$0

After Round 1 (sender): sent \$3, got back ~\$2.22, total: \$4.22

After Round 2 (receiver): received something small, total still around \$4.22

After Round 3 (sender): sent \$2, total became \$7.22 (so got ~\$0 back)

After Round 4 (receiver): received \$0.87, returned \$0.40, kept \$0.47, total: \$7.69

After Round 5 (sender): sent \$1, total became \$7.69... wait, that doesn't make sense. Let me recalculate.

Actually, let me track this differently:

Round 1 (sender): Started with \$5, sent \$3, got back \$2.22, ended with \$4.22

Round 2 (receiver): Received something... unclear

Round 3 (sender): Started with \$5, sent \$2, total after was \$7.22

Round 4 (receiver): Received \$0.87, returned \$0.40, kept \$0.47, total after was \$7.69

Round 5 (sender): Started with \$5, sent \$1, total after was \$7.69 (wait, no change?)

Round 6 (receiver): Received \$0, returned \$0, total stayed at \$12.5

Round 7 (sender): Started with \$5, sent \$2.50, total after was \$12.5 (no, it went up to \$16.28)

Round 8 (receiver): Received \$2.73, returned \$1.20, kept \$1.53, total became \$17.81

So my recent experience:

- Round 5: sent \$1, got back \$0.81 (from total change)

- Round 7: sent \$2.50, got back...

I'm getting confused about the exact mechanics here. Looking at my recent behavior as sender versus receiver, I notice I've been sending moderate amounts and getting decent reciprocation -around \$0.81 back on \$1 sent in Round 5, and a significant jump after sending \$2.50 in Round 7. When receiving, I've shown cooperation by returning roughly half of what I get, which seems to encourage future generosity....

```
Usage: {  
  "input_tokens": 1938,  
  "output_tokens": 919,  
  "reasoning_tokens": 0  
}
```

GAME - Agent_2 - Receiver - dyad_1

Timestamp: 2026-08-12T10:02:47

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {  
  "agent_type": "standard",  
  "dyad_id": "dyad_1",  
  "move_index": 1,  
  "opponent_id": "Agent_1",  
  "pairing": {  
    "agents": [  
      "Agent_1",  
      "Agent_2"  
    ],  
    "investor": "Agent_1",  
    "roles": {  
      "Agent_1": "investor",  
      "Agent_2": "trustee"  
    },  
    "trustee": "Agent_2"  
  },  
  "role": "trustee",  
  "role_label": "Receiver",  
  "round": 9,  
  "task": "game",  
  "task_index": 0  
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)
- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 3

You are the RECEIVER this round. Your total earnings across all rounds are \$7.780000000000000001.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 3 role=assistant

content:

Since the sender sent \$0 and I received \$0, there is nothing to return.

{'return': 0}

reasoning:

The sender sent me \$0 this round, so I received \$0 after the 3x multiplication.

Since I received nothing, there is nothing to return. The only valid response is to return \$0.

The format should be: {'return': 0}

This is straightforward - I can't return what I didn't receive.

usage:

```
{
  "input_tokens": 806,
  "output_tokens": 110,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 4

You are the SENDER this round. Your total earnings across all rounds are \$7.780000000000000001.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at the pattern so far:

Round 1: As receiver, the sender trusted me with 53.6% - I returned \$3.00

Round 2: As sender, I sent 54% - the receiver returned generously

Round 3: As receiver, the sender sent \$0 - possibly due to communication noise mentioned in the instructions

Given that:

1. Previous cooperation rounds were successful with good returns
2. The \$0 in round 3 might be communication noise rather than intentional
3. Maintaining cooperation has been beneficial overall

I'll continue with a cooperative strategy and send approximately 50%:

{'send': 2.5}

reasoning:

Let me analyze what happened:

Round 1: I was receiver

- Sender sent \$2.68 (53.6%)

- I received \$8.04

- I returned \$3.00

- Sender ended with \$5.32, I ended with \$5.04

Round 2: I was sender

- I sent \$2.7 (54%)

- Receiver would have received \$8.10
- My total earnings after round 2: \$7.78
- This means I started round 2 with \$5.04, sent \$2.7, so I had \$2.34 left
- My total is now \$7.78, which means the receiver returned $\$7.78 - \$2.34 = \$5.44$
- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)

...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Message 7 role=assistant

content:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

```
{'return': 0.15}
```

reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return something to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

usage:

```
{
  "input_tokens": 1099,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at the pattern of returns I've been receiving:

Round 2: I sent \$2.70, received back a small return

Round 4: I sent \$2.50, received back \$0

Despite my cooperative approach, the returns have been disappointing. This could be due to communication noise affecting what receivers see, or strategic choices by receivers.

Given this pattern, I need to be more cautious while still allowing for the possibility of re-establishing cooperation. I'll send a moderate amount to test whether cooperation can improve, but reduce my exposure.

{'send': 1.5}

reasoning:

Let me review the pattern:

Round 1: I was receiver. Sender sent \$2.68 (53.6%), I received \$8.04, I returned \$3.00

Round 2: I was sender. I sent \$2.70 (54%), receiver presumably returned well

Round 3: I was receiver. Sender sent \$0 (communication noise?)

Round 4: I was sender. I sent \$2.50 (50%)

Round 5: I was receiver. Sender sent \$0.13 (2.6%), I received \$0.39, I returned \$0.15

My total earnings are \$10.52, which means:

- Round 1: I got \$5.04 (started with \$5, sent \$2.68, got \$3.00 back would be \$5.32... wait that doesn't match)

Let me recalculate from the receiver perspective in Round 1:

- I received \$8.04
- I returned \$3.00

- I kept \$5.04
So my earnings from Round 1 were \$5.04

Round 2: I was sender with \$5 endowment

- I sent \$2.70
- I had \$2.30 left
- The receiver must have returned something to get me to \$7.78 total

$\$7.78 - \$5.04 = \$2.74$ earnings in round 2

If I sent \$2.70 and kept \$2.30, then I received back... wait

Actually as sender in round 2, I start fresh with \$5

- I sent \$2.70
- I kept \$2.30
- Receiver returned X
- My total from round 2 = $\$2.30 + X$

Total after round 2 was \$7.78

So the receiver returned only \$0.44 on the \$8.10 they received-surprisingly low given what I sent. Round 3 was a wash with the sender contributing nothing. In Round 4, I sent \$2.50 and kept the other half, earning exactly that amount back with no return, suggesting the receiver kept everything. Round 5 shows minimal reciprocation again: the sender only sent \$0.13 (2.6% of the tripled amount), and I returned \$0.15 on my \$0.39 received.

I'm noticing a troubling pattern-my generous sends in Rounds 2 and 4 aren't being matched with proportional returns, while I'm getting minimal amounts as a receiver. This could indicate either communication issues distorting the amounts or a lack of reciprocal trust from the other players....

usage:

```
{
  "input_tokens": 1307,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.160000000000002.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 11 role=assistant

content:

The sender sent \$0.29 (5.8% of their endowment), which is relatively low. This might be due to communication noise or decreased trust after previous rounds.

I'll continue to maintain proportional fairness in my returns. Returning approximately 40% of what I received:

If I return \$0.35:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains a fair reciprocal relationship.

```
{'return': 0.35}
```

reasoning:

The sender sent \$0.29 (5.8% of their endowment), which is quite low. Looking at the pattern:

Round 5: Sender sent \$0.13, I received \$0.39 and returned \$0.15

Round 7: Sender sent \$0.29, I received \$0.87

There seems to be a pattern of very low sends lately. This could be:

1. Communication noise (the sender might be sending more than I see)
2. The sender becoming less trusting after poor returns
3. Strategic play

Given that I've tried to maintain fairness in my returns, I should continue to return a proportional amount. If I return approximately 40% of what I received:

$\$0.87 \times 0.4 = \0.35

This would give:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains the reciprocal fairness approach.

usage:

```
{
  "input_tokens": 1514,
  "output_tokens": 385,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.680000000000001.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Reviewing my experience as sender:

Round 2: Sent \$2.70 (cooperative) - received some return

Round 4: Sent \$2.50 (cooperative) - received \$0 back

Round 6: Sent \$1.50 (reduced after poor returns) - outcome unclear

The pattern shows that when I send larger amounts, returns have been disappointing. This could be due to communication noise affecting what receivers perceive, or other factors.

I'll continue with a cautious approach, sending a small amount to minimize losses while still allowing for potential cooperation:

{'send': 1.0}

reasoning:

Let me review what has happened as sender:

Round 2: I sent \$2.70, and the receiver returned a small amount (looks like from my earnings change)

Round 4: I sent \$2.50, and the receiver returned \$0

Round 6: I sent \$1.50, and the receiver returned... let me check

My earnings trajectory:

- Start Round 2: \$5.04
- After Round 2 (sent \$2.70): Should have \$2.34 + whatever was returned
- Round 3: \$7.78 (so I gained \$2.74 from Round 2 return, meaning I got back more than I sent!)
- After Round 4 (sent \$2.50): Start was \$7.78, ended at \$7.78 (so got nothing back)
- Round 6: Started at \$10.52
- After Round 6 (sent \$1.50): Should have \$3.50 + whatever was returned
- Round 8 start: \$14.68

Wait, let me recalculate:

Round 6 start: \$10.52

Sent \$1.50, kept \$3.50

Round 8 start: \$14.68

Difference: $\$14.68 - \$10.52 = \$4.16$

Net from Round 6: \$4.16

I sent \$1.50, so net change = $-\$1.50 + \text{return}$

$\$4.16 = -\$1.50 + \text{return} + \text{earnings as receiver in rounds 5 and 7}$

Actually, I need to track this differently. Let me look at what I know:

- Round 5 and 7: I was receiver
- Round 5: Received \$0.39, returned \$0.15, kept \$0.24
- Round 7: Received \$0.87, returned \$0.35, kept \$0.52

So from being receiver in rounds 5 and 7, I gained \$0.76 total. Looking at my overall trajectory from round 3 to round 8, I went from \$7.78 to \$14.68, a gain of \$6.90. Subtracting

g what I earned as receiver (\$0.76), that leaves \$6.14 from my sender rounds. In round 4, I sent \$2.50 and got nothing back, so the return must have come from round 6....

usage:

```
{
  "input_tokens": 1448,
  "output_tokens": 839,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.32.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Assistant output:

Since the sender sent \$0 and I received \$0, there is nothing to return.

```
{'return': 0}
```

Reasoning:

The sender sent \$0 and I received \$0. There's nothing to return since I received nothing.

This is straightforward - with \$0 received, I can only return \$0.

Usage: {

```
"input_tokens": 1490,
"output_tokens": 78,
"reasoning_tokens": 0
}
```

ROUND 10

Roles: Agent_1=Receiver, Agent_2=Sender

Pairings:

dyad_1: Agent_2=Sender, Agent_1=Receiver

Parsed game decisions:

dyad_1: Agent_2 sent \$1.5; received \$4.5; Agent_1 returned \$0.7; payoffs Agent_2=\$4.2, Agent_1=\$3.8; noise/visible: sent visible=\$0.53, received visible=\$1.59, returned visible=\$1.14
Action validation: {"returned": {"amount": 0.7, "clamped": false, "max": 4.5, "min": 0, "raw": 0.7}, "sent": {"amount": 1.5, "clamped": false, "max": 5, "min": 0, "raw": 1.5}}

Balances after round: {

```
"Agent_1": 42.3,
"Agent_2": 49.10000000000001
}
```

Visible balances after round: {

```
"Agent_1": 20.700000000000003,
"Agent_2": 24.96
}
```

LLM CALLS

GAME - Agent_2 - Sender - dyad_1

Timestamp: 2026-08-12T10:02:58

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

Metadata: {

```
"agent_type": "standard",
"dyad_id": "dyad_1",
"move_index": 0,
"opponent_id": "Agent_1",
"pairing": {
  "agents": [
    "Agent_2",
    "Agent_1"
  ],
"investor": "Agent_2",
"roles": {
  "Agent_1": "trustee",
  "Agent_2": "investor"
},
}
```

Exact input messages sent to model:

content:

GAME :

Let me analyze what happened:

Round 1: I was receiver

- Sender sent \$2.68 (53.6%)
- I received \$8.04
- I returned \$3.00
- Sender ended with \$5.32, I ended with \$5.04

Round 2: I was sender

- I sent \$2.7 (54%)
- Receiver would have received \$8.10
- My total earnings after round 2: \$7.78
- This means I started round 2 with \$5.04, sent \$2.7, so I had \$2.34 left
- My total is now \$7.78, which means the receiver returned $\$7.78 - \$2.34 = \$5.44$
- That's a very generous return! (\$5.44 out of \$8.10 received)

Round 3: I was receiver

- Sender sent \$0
- I received \$0
- I returned \$0
- This is interesting - why would they send \$0?

Now Round 4: I'm sender again

- I have \$5 to potentially send

The pattern so far:

- Round 1: Sender sent 53.6%, I returned reasonably (37% of what I received)
- Round 2: I sent 54%, receiver returned generously (67% of what they received)
- Round 3: Sender sent 0%

The \$0 send in round 3 is puzzling. It could be:

1. They're testing me
2. Communication noise (though \$0 seems unlikely to be noise)
3. They decided not to trust me for some reason
4. Strategic decision

But wait - there's "communication noise" mentioned in the instructions. The amounts I see may differ from what was actually sent. Maybe the sender in round 3 actually intended to send something, but noise made it appear as \$0?

Given:

- The previous cooperation was very positive (receiver returned generously in round 2)
- ...

usage:

```
{
  "input_tokens": 883,
  "output_tokens": 674,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the RECEIVER this round. Your total earnings across all rounds are \$10.280000000000001.

This round, The sender sent you \$0.13, so you received \$0.39. How much do you return to The sender? (0-0.39)

Message 5 role=assistant

content:

The sender sent only \$0.13 (2.6% of their endowment), which is much lower than previous rounds. This could be due to the communication noise mentioned in the instructions, or a strategic choice.

Even though the amount is small, I should acknowledge the trust shown and return a fair proportion to maintain the cooperative relationship.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$

This is roughly a 38% return rate, which matches the proportional fairness from earlier rounds.

```
{'return': 0.15}
```

reasoning:

The sender sent \$0.13, which is only 2.6% of their endowment. This is a dramatic drop from the ~50% we were seeing earlier.

I received \$0.39 after the 3x multiplication.

This could be:

1. Communication noise (as mentioned in the instructions)
2. A strategic choice by the sender
3. A test of some kind

Even though the amount is very small, the sender did show some trust. I should return something to maintain the relationship, but given the small amount sent, a proportional return would make sense.

If I return \$0.15:

- Sender ends with: $\$4.87 + \$0.15 = \$5.02$
- I end with: $\$0.39 - \$0.15 = \$0.24$
- The sender gets a small gain, I get a small gain

This rewards the trust shown (small as it is) while being proportional to the amount sent.

Alternatively, I could return a bit more to signal willingness to cooperate, or less since the amount sent was so small.

Let me think about what's fair:

- Sender sent 2.6% of endowment
- If they had sent 50% like before, I would have returned maybe \$3-3.5 out of \$7.50
- That's about 40-47% return
- 40% of \$0.39 = \$0.156

I'll return \$0.15 to maintain fairness and signal continued willingness to cooperate.

usage:

```
{
  "input_tokens": 1099,
  "output_tokens": 513,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the SENDER this round. Your total earnings across all rounds are \$10.520000000000001.

How much do you send to the receiver this round? (0-5)

Message 7 role=assistant

content:

Looking at the pattern of returns I've been receiving:

Round 2: I sent \$2.70, received back a small return

Round 4: I sent \$2.50, received back \$0

Despite my cooperative approach, the returns have been disappointing. This could be due to communication noise affecting what receivers see, or strategic choices by receivers.

Given this pattern, I need to be more cautious while still allowing for the possibility of re-establishing cooperation. I'll send a moderate amount to test whether cooperation can improve, but reduce my exposure.

```
{'send': 1.5}
```

reasoning:

Let me review the pattern:

Round 1: I was receiver. Sender sent \$2.68 (53.6%), I received \$8.04, I returned \$3.00

Round 2: I was sender. I sent \$2.70 (54%), receiver presumably returned well

Round 3: I was receiver. Sender sent \$0 (communication noise?)

Round 4: I was sender. I sent \$2.50 (50%)

Round 5: I was receiver. Sender sent \$0.13 (2.6%), I received \$0.39, I returned \$0.15

My total earnings are \$10.52, which means:

- Round 1: I got \$5.04 (started with \$5, sent \$2.68, got \$3.00 back would be \$5.32... wait that doesn't match)

Let me recalculate from the receiver perspective in Round 1:

- I received \$8.04
- I returned \$3.00
- I kept \$5.04

So my earnings from Round 1 were \$5.04

Round 2: I was sender with \$5 endowment

- I sent \$2.70
- I had \$2.30 left
- The receiver must have returned something to get me to \$7.78 total

$\$7.78 - \$5.04 = \$2.74$ earnings in round 2

If I sent \$2.70 and kept \$2.30, then I received back... wait

Actually as sender in round 2, I start fresh with \$5

- I sent \$2.70
- I kept \$2.30
- Receiver returned X
- My total from round 2 = $\$2.30 + X$

Total after round 2 was \$7.78

So the receiver returned only \$0.44 on the \$8.10 they received-surprisingly low given what I sent. Round 3 was a wash with the sender contributing nothing. In Round 4, I sent \$2.50 and kept the other half, earning exactly that amount back with no return, suggesting the receiver kept everything. Round 5 shows minimal reciprocation again: the sender only sent \$0.13 (2.6% of the tripled amount), and I returned \$0.15 on my \$0.39 received.

I'm noticing a troubling pattern-my generous sends in Rounds 2 and 4 aren't being matched with proportional returns, while I'm getting minimal amounts as a receiver. This could indicate either communication issues distorting the amounts or a lack of reciprocal trust from the other players....

usage:

```
{
  "input_tokens": 1307,
  "output_tokens": 1049,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the RECEIVER this round. Your total earnings across all rounds are \$14.160000000000002.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 9 role=assistant

content:

The sender sent \$0.29 (5.8% of their endowment), which is relatively low. This might be due to communication noise or decreased trust after previous rounds.

I'll continue to maintain proportional fairness in my returns. Returning approximately 40% of what I received:

If I return \$0.35:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains a fair reciprocal relationship.

{'return': 0.35}

reasoning:

The sender sent \$0.29 (5.8% of their endowment), which is quite low. Looking at the pattern:

Round 5: Sender sent \$0.13, I received \$0.39 and returned \$0.15
Round 7: Sender sent \$0.29, I received \$0.87

There seems to be a pattern of very low sends lately. This could be:

1. Communication noise (the sender might be sending more than I see)
2. The sender becoming less trusting after poor returns
3. Strategic play

Given that I've tried to maintain fairness in my returns, I should continue to return a proportional amount. If I return approximately 40% of what I received:

$$\$0.87 \times 0.4 = \$0.35$$

This would give:

- Sender ends with: $\$4.71 + \$0.35 = \$5.06$
- I end with: $\$0.87 - \$0.35 = \$0.52$

This maintains the reciprocal fairness approach.

usage:

```
{  
  "input_tokens": 1514,  
  "output_tokens": 385,  
  "reasoning_tokens": 0  
}
```

Message 10 role=user

content:

Round 8

You are the SENDER this round. Your total earnings across all rounds are \$14.680000000000001.

How much do you send to the receiver this round? (0-5)

Message 11 role=assistant

content:

Reviewing my experience as sender:

Round 2: Sent \$2.70 (cooperative) - received some return

Round 4: Sent \$2.50 (cooperative) - received \$0 back

Round 6: Sent \$1.50 (reduced after poor returns) - outcome unclear

The pattern shows that when I send larger amounts, returns have been disappointing. This could be due to communication noise affecting what receivers perceive, or other factors.

I'll continue with a cautious approach, sending a small amount to minimize losses while still allowing for potential cooperation:

```
{'send': 1.0}
```

reasoning:

Let me review what has happened as sender:

Round 2: I sent \$2.70, and the receiver returned a small amount (looks like from my earnings change)

Round 4: I sent \$2.50, and the receiver returned \$0

Round 6: I sent \$1.50, and the receiver returned... let me check

My earnings trajectory:

- Start Round 2: \$5.04
- After Round 2 (sent \$2.70): Should have \$2.34 + whatever was returned
- Round 3: \$7.78 (so I gained \$2.74 from Round 2 return, meaning I got back more than I sent!)
- After Round 4 (sent \$2.50): Start was \$7.78, ended at \$7.78 (so got nothing back)
- Round 6: Started at \$10.52
- After Round 6 (sent \$1.50): Should have \$3.50 + whatever was returned
- Round 8 start: \$14.68

Wait, let me recalculate:

Round 6 start: \$10.52

Sent \$1.50, kept \$3.50

Round 8 start: \$14.68

Difference: $\$14.68 - \$10.52 = \$4.16$

Net from Round 6: \$4.16

I sent \$1.50, so net change = $-\$1.50 + \text{return}$
 $\$4.16 = -\$1.50 + \text{return} + \text{earnings as receiver in rounds 5 and 7}$

Actually, I need to track this differently. Let me look at what I know:

- Round 5 and 7: I was receiver
- Round 5: Received \$0.39, returned \$0.15, kept \$0.24
- Round 7: Received \$0.87, returned \$0.35, kept \$0.52

So from being receiver in rounds 5 and 7, I gained \$0.76 total. Looking at my overall trajectory from round 3 to round 8, I went from \$7.78 to \$14.68, a gain of \$6.90. Subtracting what I earned as receiver (\$0.76), that leaves \$6.14 from my sender rounds. In round 4, I sent \$2.50 and got nothing back, so the return must have come from round 6....

usage:

```
{
  "input_tokens": 1448,
  "output_tokens": 839,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the RECEIVER this round. Your total earnings across all rounds are \$20.32.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 13 role=assistant

content:

Since the sender sent \$0 and I received \$0, there is nothing to return.

```
{'return': 0}
```

reasoning:

The sender sent \$0 and I received \$0. There's nothing to return since I received nothing.

This is straightforward - with \$0 received, I can only return \$0.

usage:

```
{
  "input_tokens": 1490,
  "output_tokens": 78,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the SENDER this round. Your total earnings across all rounds are \$20.32.

How much do you send to the receiver this round? (0-5)

Assistant output:

Looking at my experience across all rounds:

As SENDER:

- Round 2: Sent \$2.70 (cooperative)
- Round 4: Sent \$2.50 (cooperative)
- Round 6: Sent \$1.50 (reduced caution)
- Round 8: Sent \$1.00 (cautious)

The returns have generally been disappointing, though this could be due to communication noise.

This is Round 10, likely the final round. While there's no future cooperation to maintain, I still want to demonstrate some willingness to participate fairly. However, given the pattern of poor returns, I'll be moderately cautious.

```
{'send': 1.5}
```

Reasoning:

Let me review the pattern as sender across rounds:

- Round 2: Sent \$2.70 - cooperative approach
- Round 4: Sent \$2.50 - maintaining cooperation
- Round 6: Sent \$1.50 - reduced after poor returns
- Round 8: Sent \$1.00 - cautious approach

The returns I've received have generally been disappointing. This could be due to:

1. Communication noise affecting what receivers see
2. Strategic choices by receivers
3. A combination of both

My total earnings are \$20.32 across all rounds, which suggests I haven't been losing too much, but cooperation hasn't been very rewarding either.

For this final round (Round 10), I need to consider:

- This is the last round, so there's no future cooperation to maintain
- However, I still want to be somewhat fair
- Given the poor returns in previous rounds, sending a large amount seems risky
- But sending nothing seems too defensive

I think I'll send a moderate amount - maybe \$2.00 - which shows some willingness to cooperate but protects most of my endowment. This balances fairness with self-protection given the pattern of returns.

Actually, looking at my earnings of \$20.32 total, let me think about what makes sense. I've played 10 rounds, so average per round is about \$2.03. That's below the \$5 endowment when I'm sender, suggesting my sending strategy has been costly.

Given it's the last round and past returns have been poor, I should be more conservative. I'll send \$1.50.

```
Usage: {
  "input_tokens": 1472,
  "output_tokens": 527,
  "reasoning_tokens": 0
}
```

GAME - Agent_1 - Receiver - dyad_1

Timestamp: 2026-08-12T10:03:06

Model: anthropic/claude-sonnet-4.5

Temperature: 0.8

```
Metadata: {
  "agent_type": "standard",
  "dyad_id": "dyad_1",
  "move_index": 1,
  "opponent_id": "Agent_2",
  "pairing": {
    "agents": [
      "Agent_2",
      "Agent_1"
    ],
    "investor": "Agent_2",
    "roles": {
      "Agent_1": "trustee",
      "Agent_2": "investor"
    },
    "trustee": "Agent_1"
  },
  "role": "trustee",
  "role_label": "Receiver",
  "round": 10,
  "task": "game",
  "task_index": 0
}
```

Exact input messages sent to model:

Message 1 role=system

content:

You are an agent participating in a multi-task experiment. You will play a game and may also be asked to write myths.

GAME:

You will play different roles across multiple rounds.

When you are the SENDER:

- You receive \$5 each round
- You decide how much to SEND to the RECEIVER (\$0-\$5)

- Amount sent is multiplied by 3x
- Receiver decides how much to RETURN
- You keep: \$5 - sent + returned
- Receiver keeps: received - returned
- Response format: {'send': <amount>}

When you are the RECEIVER:

- Sender receives \$5
- Sender decides how much to SEND to you (\$0-\$5)
- Amount sent is multiplied by 3x
- You decide how much to RETURN
- Sender keeps: \$5 - sent + returned
- You keep: received - returned
- Response format: {'return': <amount>}

MYTH WRITING:

You may also be asked to write short myths (stories) on various topics. When writing myths, follow the instructions provided in each prompt.

IMPORTANT: Provide your decision in the correct JSON format.

NOTE: There may be communication noise in this game.

The amounts you see may differ slightly from what was actually sent/returned.

This is part of the experimental design.

Message 2 role=user

content:

Round 4

You are the RECEIVER this round. Your total earnings across all rounds are \$7.220000000000001.

This round, The sender sent you \$0.29, so you received \$0.8699999999999999. How much do you return to The sender? (0-0.8699999999999999)

Message 3 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.29, which became \$0.87 after the 3x multiplier
- This is a very small amount - possibly due to communication noise or low trust
- My total earnings show that cooperation hasn't been highly rewarding so far

The sender took a small risk by sending anything at all. To encourage future cooperation and signal that I'm willing to reciprocate fairly, I'll return a reasonable portion:

```
'''json
{"return": 0.40}
'''
```

This returns approximately 46% of what I received, which:

- Gives the sender back more than they sent (\$0.40 vs \$0.29)
- Shows I'm cooperative even with small amounts
- Keeps some value from the interaction (\$0.47 for me)
- Might help rebuild trust for future rounds despite the communication noise issues

reasoning:

Let me analyze the situation:

Round 1 (SENDER): I sent \$3, my total after was \$4.22, so I received back \$2.22

- I kept: \$5 - \$3 + \$2.22 = \$4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- My earnings stayed at \$4.22

Round 3 (SENDER): I sent \$2, my total is now \$7.22

- Change from \$4.22 to \$7.22 = \$3
- I kept: \$5 - \$2 + returned = \$7.22
- So: \$3 + returned = \$7.22
- Returned = \$4.22

Wait, that doesn't make sense. Let me recalculate.

Actually, the total earnings are cumulative across all rounds.

Round 1 (SENDER): I sent \$3

- If I received back \$2.22, I kept: $\$5 - \$3 + \$2.22 = \4.22 ?

Round 2 (RECEIVER): I received \$0, returned \$0

- I gained: $\$0 - \$0 = \$0$

- Total: $\$4.22 + \$0 = \$4.22$?

Round 3 (SENDER): I sent \$2

- Total before Round 3: \$4.22

- Total after Round 3: \$7.22

- Change: \$3.00

- I kept: $\$5 - \$2 + \text{returned} = \$3$

- So returned = \$0

Hmm, so in Round 3, the receiver returned \$0 to me? That's unfortunate.

Round 4 (RECEIVER): I received \$0.87

- The sender sent \$0.29, which became \$0.87 after 3x multiplier

- This is a minimal amount, possibly testing my willingness to reciprocate or signaling low confidence in...

usage:

```
{
  "input_tokens": 1098,
  "output_tokens": 843,
  "reasoning_tokens": 0
}
```

Message 4 role=user

content:

Round 5

You are the SENDER this round. Your total earnings across all rounds are \$7.69.

How much do you send to the receiver this round? (0-5)

Message 5 role=assistant

content:

Looking at my earnings progression:

****Analysis:****

- Round 1: Sent \$3, got back ~\$2.22 (receiver kept ~75%)

- Round 3: Sent \$2, got back \$0 (receiver kept everything!)

- Round 4: As receiver, I returned \$0.40 from \$0.87 received

The pattern shows receivers have been largely uncooperative with me, particularly in Round 3 where I got nothing back. However:

- Communication noise may be affecting outcomes

- Each round might have different partners

- Complete mistrust (sending \$0) guarantees no gains

I'll send a minimal amount to test cooperation while protecting against further losses:

```
'''json
{"send": 1}
'''
```

This sends \$1, which becomes \$3 for the receiver. If they return even \$1-1.50, we both gain. If they keep everything again, I've limited my loss. This cautious approach reflects the poor reciprocation history while still leaving room for potential cooperation.

reasoning:

Let me track my earnings history:

Round 1: I was SENDER, sent \$3

- Started with \$5

- Sent \$3, so I had \$2 left

- My total after Round 1 was \$4.22

- So I received back: $\$4.22 - \$2 = \$2.22$

Round 2: I was RECEIVER, received \$0

- No change to my earnings except what I already had

Round 3: I was SENDER, sent \$2

- I had \$4.22 before this round
- Started this round with \$5, sent \$2, so I had \$3 left
- My total after Round 3 should have been $\$4.22 + \$3 + (\text{what was returned})$
- After Round 3, my total was \$7.22
- So: $\$4.22 + \$3 + \text{returned} = \$7.22$
- $\text{Returned} = \$7.22 - \$4.22 - \$3 = \0
- Wait, that doesn't work...
- Actually, let me recalculate. After Round 2, I had \$4.22
- In Round 3, I start fresh with \$5, send \$2, keep \$3
- So I have \$3 + whatever is returned
- After Round 3, my TOTAL is \$7.22
- So $\$4.22 (\text{previous}) + \$3 (\text{kept in Round 3}) + \text{returned} = \7.22
- $\text{Returned} = \$0$

So in Round 3, the receiver didn't return anything!

Round 4: I was RECEIVER, received \$0.87

- I returned \$0.40
- So I kept $\$0.87 - \$0.40 = \$0.47$
- My total before was \$7.22, now should be $\$7.22 + \$0.47 = \$7.69$?
- When I sent \$3 in Round 1, it tripled to \$9, but the receiver only gave back \$2.22, keeping about 75%. In Round 3, I sent \$2 which became \$6, yet the receiver returned nothing at all. The people I've been paired with haven't shown much generosity-one kept three-quarters of what I sent, the other kept it all....

usage:

```
{
  "input_tokens": 1340,
  "output_tokens": 902,
  "reasoning_tokens": 0
}
```

Message 6 role=user

content:

Round 6

You are the RECEIVER this round. Your total earnings across all rounds are \$12.5.

This round, The sender sent you \$0, so you received \$0. How much do you return to The sender? (0-0)

Message 7 role=assistant

content:

I understand the situation:

- The sender sent me \$0 this round
- After the 3x multiplier, I received \$0
- I have nothing to return

```
'''json
{"return": 0}
'''
```

****Observation:**** This is the second time as a receiver that I've been sent nothing. The lack of sending suggests either very low trust in the system, previous bad experiences with receivers, or the effects of communication noise discouraging cooperation. It's unfortunate as the 3x multiplier means we're both missing out on potential gains.

reasoning:

The sender sent me \$0 this round, so after the 3x multiplier I received \$0.

Since I received nothing, I can only return \$0.

My total earnings are now \$12.50, which means in Round 5, I must have gained \$4.81 (from \$7.69 to \$12.50). This suggests:

- I sent \$1 in Round 5
- I should have kept \$4 if I sent \$1
- But I gained \$4.81, which means I got back approximately \$0.81

Wait, let me recalculate:

- Round 4 end: \$7.69
- Round 5: I was sender, sent \$1
- Started with \$5

- Sent \$1, so kept \$4
- Plus whatever was returned
- Total gain = $\$12.5 - \$7.69 = \$4.81$
- So I must have been returned \$0.81
- That means: $\$5 - \$1 + \$0.81 = \4.81 ?

The receiver in Round 5 kept most of the \$3 they received (returning only \$0.81 out of \$3)

Now in Round 6, I'm the receiver and got \$0, so I return \$0.

usage:

```
{
  "input_tokens": 1628,
  "output_tokens": 431,
  "reasoning_tokens": 0
}
```

Message 8 role=user

content:

Round 7

You are the SENDER this round. Your total earnings across all rounds are \$12.5.

How much do you send to the receiver this round? (0-5)

Message 9 role=assistant

content:

Looking at my earnings history:

****Analysis:****

- Round 1: Sent \$3, got back \$2.22 (modest return)
- Round 3: Sent \$2, got back \$0 (poor cooperation)
- Round 5: Sent \$1, got back \$0.81 (good reciprocation!)

The most recent round where I was sender (Round 5) showed encouraging cooperation - I got back nearly as much as I sent, despite the receiver keeping most of the tripled amount.

This suggests either:

- Trust is building
- I encountered a more cooperative partner
- The lower stakes made reciprocation easier

Given this positive trend and my comfortable position at \$12.50, I'll send a moderate amount to test continued cooperation:

```
'''json
{"send": 2.5}
'''
```

This sends \$2.50, which becomes \$7.50 for the receiver. If they return \$2-3, we both benefit significantly. The improving pattern suggests this might be worthwhile, while the amount protects me if cooperation breaks down again.

reasoning:

Let me track my earnings progression:

- Start Round 1: \$0
- Round 1 (sender): Sent \$3, so I have $\$5 - \$3 = \$2$ at first. Received back ~\$2.22. Total: \$4.22
- Round 2 (receiver): Got \$0. Total stays: \$4.22
- Round 3 (sender): Sent \$2, so spent \$2. Total before return: $\$4.22 - \$2 + \$5 = \7.22 . Then got back... let me calculate. My total was \$7.22 after Round 3.
- Round 4 (receiver): Got \$0.87, returned \$0.40. Kept \$0.47. Total: $\$7.22 + \$0.47 = \$7.69$
- Round 5 (sender): Sent \$1, started with \$5. So $\$5 - \$1 = \$4$. Added to \$7.69 = \$11.69. Then got back... My total was \$7.69 after Round 5 per the game. Wait, that doesn't make sense.

Let me reconsider. The game tells me my total earnings across all rounds at the start of each round.

Round 1 start: I'm sender, have \$5

Round 2 start: Total = \$4.22 (so in Round 1, I ended with \$4.22)

Round 3 start: Total = \$4.22 (so Round 2 I gained \$0)

Round 4 start: Total = \$7.22 (so Round 3 I gained \$3)

Round 5 start: Total = \$7.69 (so Round 4 I gained \$0.47)

Round 6 start: Total = \$12.5 (so Round 5 I gained \$4.81)

Round 7 start: Total = \$12.5 (so Round 6 I gained \$0)

Now let me verify each round's mechanics. Round 1 as sender: I sent \$3 and got back \$2.22, netting \$4.22 from my initial \$5. Round 2 receiving nothing meant no change. Round 3 I sent \$2 but the math isn't adding up-I need to recalculate what I actually received back in that round.

Looking at the total progression, Round 3 went from \$4.22 to \$7.22, a \$3 gain, which means I got \$0 back despite sending \$2. Round 4 as receiver brought in \$0.87 and I returned \$0.40, adding \$0.47. Round 5 as sender shows my total jumped from \$7.69 to \$12.5-a \$4.81 gain-which means I sent \$1 but received \$0.81 back. Round 6 receiving \$0 and returning \$0 produced no change.

The returns I've gotten as a sender have been inconsistent: \$2.22 on my first...

usage:

```
{
  "input_tokens": 1795,
  "output_tokens": 1286,
  "reasoning_tokens": 0
}
```

Message 10 role=user

content:

Round 8

You are the RECEIVER this round. Your total earnings across all rounds are \$16.28.

This round, The sender sent you \$0.91, so you received \$2.73. How much do you return to the sender? (0-2.73)

Message 11 role=assistant

content:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- This is a small amount, suggesting caution or low trust
- In Round 4, I showed cooperative behavior by returning more than the sender sent

To maintain a reputation for fair reciprocation and encourage future cooperation, I'll return a reasonable portion that rewards the sender's trust:

```
'''json
{"return": 1.20}
'''
```

This returns approximately 44% of what I received, which:

- Gives the sender back significantly more than they sent (\$1.20 vs \$0.91)
- Net gain for sender: about \$0.29
- Net gain for me: \$1.53
- Demonstrates I'm cooperative and trustworthy
- Should encourage larger sends in future rounds if we interact again

reasoning:

Let me analyze the situation:

****Analysis:****

- The sender sent me \$0.91, which became \$2.73 after the 3x multiplier
- My total earnings are \$16.28, which shows I've been doing reasonably well
- This is a relatively small amount sent, suggesting either low trust or the effects of communication noise

Looking at my previous behavior:

- Round 4: I received \$0.87 and returned \$0.40 (about 46% of what I received, more than sender sent)
- This showed I'm cooperative and willing to reward trust

The sender sent a small amount (\$0.91), which suggests they're being cautious. To encourage cooperation and show I'm trustworthy, I should return a fair amount.

If I return around 40-50% of what I received:

- 40% of \$2.73 = \$1.09
- 50% of \$2.73 = \$1.37

Returning about \$1.20-\$1.30 would:

- Give the sender back more than they sent (\$0.91)
- Show I'm reciprocating fairly
- Encourage future higher sends
- Keep a reasonable amount for myself (\$1.43-\$1.53)

I'll return \$1.20 as it's a good balance - clearly more than the sender sent, showing I reward trust, but not so much that I'm giving away everything.

usage:

```
{
  "input_tokens": 1898,
  "output_tokens": 522,
  "reasoning_tokens": 0
}
```

Message 12 role=user

content:

Round 9

You are the SENDER this round. Your total earnings across all rounds are \$17.810000000000002.

How much do you send to the receiver this round? (0-5)

Message 13 role=assistant

content:

Looking at my earnings and recent trends:

****Analysis:****

- My total earnings have grown from \$12.50 to \$17.81
- Round 5 (sender): Sent \$1, received good reciprocation
- Round 7 (sender): Sent \$2.50, contributed to earnings growth
- Round 8 (receiver): Returned \$1.20 from \$2.73, showing I'm cooperative

The pattern shows improving cooperation in the game. My strong earnings position (\$17.81) allows me to take a reasonable risk to maximize potential gains while signaling trust:

```
'''json
{"send": 3}
'''
```

This sends \$3, which becomes \$9 for the receiver. If they reciprocate at a similar rate to what I've been doing (40-45%), I'll get back \$3.60-4.05, giving us both solid gains. The amount is substantial enough to show trust while my overall position can absorb it if cooperation fails.

reasoning:

Let me analyze my earnings progression:

Starting: \$0

After Round 1 (sender): sent \$3, got back ~\$2.22, total: \$4.22

After Round 2 (receiver): received something small, total still around \$4.22

After Round 3 (sender): sent \$2, total became \$7.22 (so got ~\$0 back)

After Round 4 (receiver): received \$0.87, returned \$0.40, kept \$0.47, total: \$7.69

After Round 5 (sender): sent \$1, total became \$7.69... wait, that doesn't make sense. Let me recalculate.

Actually, let me track this differently:

Round 1 (sender): Started with \$5, sent \$3, got back \$2.22, ended with \$4.22

Round 2 (receiver): Received something... unclear

Round 3 (sender): Started with \$5, sent \$2, total after was \$7.22

Round 4 (receiver): Received \$0.87, returned \$0.40, kept \$0.47, total after was \$7.69

Round 5 (sender): Started with \$5, sent \$1, total after was \$7.69 (wait, no change?)

Round 6 (receiver): Received \$0, returned \$0, total stayed at \$12.5

Round 7 (sender): Started with \$5, sent \$2.50, total after was \$12.5 (no, it went up to \$16.28)

Round 8 (receiver): Received \$2.73, returned \$1.20, kept \$1.53, total became \$17.81

So my recent experience:

- Round 5: sent \$1, got back \$0.81 (from total change)
- Round 7: sent \$2.50, got back...

I'm getting confused about the exact mechanics here. Looking at my recent behavior as send

er versus receiver, I notice I've been sending moderate amounts and getting decent reciprocation-around \$0.81 back on \$1 sent in Round 5, and a significant jump after sending \$2.50 in Round 7. When receiving, I've shown cooperation by returning roughly half of what I get, which seems to encourage future generosity....

usage:

```
{
  "input_tokens": 1938,
  "output_tokens": 919,
  "reasoning_tokens": 0
}
```

Message 14 role=user

content:

Round 10

You are the RECEIVER this round. Your total earnings across all rounds are \$19.810000000000002.

This round, The sender sent you \$0.53, so you received \$1.59. How much do you return to The sender? (0-1.59)

Assistant output:

Looking at the situation:

****Analysis:****

- The sender sent me \$0.53, which became \$1.59 after the 3x multiplier
- This is another small send, suggesting continued caution in the game
- My earnings are strong at \$19.81
- I've been consistent in returning roughly 44-46% as a receiver

Maintaining my cooperative pattern:

```
'''json
{"return": 0.70}
'''
```

This returns approximately 44% of what I received, which:

- Gives the sender back more than they sent (\$0.70 vs \$0.53)
- Net gain for sender: about \$0.17
- Net gain for me: \$0.89
- Keeps me consistent with my reciprocity pattern
- Signals reliability even with small amounts

Reasoning:

Looking at the situation:

- The sender sent me \$0.53, which became \$1.59 after the 3x multiplier
- This is a small amount again
- My total earnings are \$19.81, so I'm doing well
- In previous rounds as receiver (Round 4 and Round 8), I've shown cooperative behavior by returning more than what was sent

Let me maintain consistency with my cooperative strategy:

- Round 4: Returned \$0.40 from \$0.87 (46%)
- Round 8: Returned \$1.20 from \$2.73 (44%)

I should return a similar percentage (around 40-45%) to maintain my reputation as a fair player.

$\$1.59 \times 0.44 = \0.70 approximately

This would give the sender back more than they sent (\$0.70 vs \$0.53), showing I'm trustworthy even with small amounts.

Usage:

```
{
  "input_tokens": 1945,
  "output_tokens": 401,
  "reasoning_tokens": 0
}
```